COMP0087

Statistical Natural Language Processing

Lecture 11 – Reasoning

What is reasoning?

Logic: The process of deriving conclusions from premises according to valid rules of inference.
Cognitive Science: The cognitive process by which an agent draws inferences, forms judgments, or updates beliefs.
Symbolic AI: Algorithmic manipulation of symbolic representations to produce new representations consistent with a knowledge base.

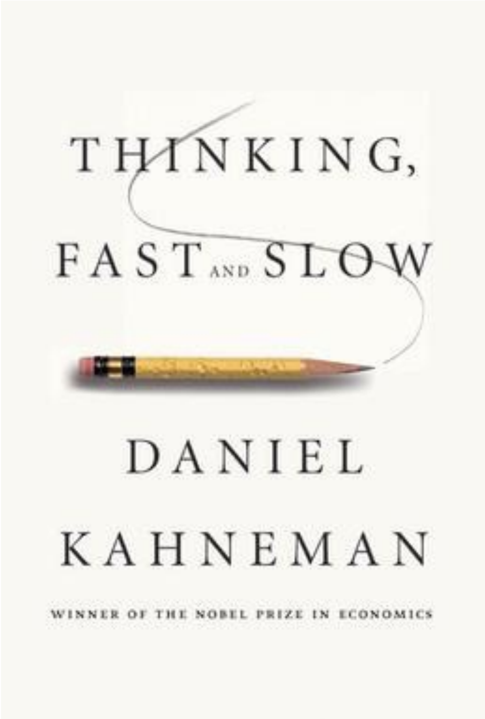
Is Socrates mortal?
Socrates is a man.
Men are mortal.
Yes, Socrates is mortal.

3x-1=2. What is x?
(3x-1)+1=2+1
3x=3
3x/3=3/3
x=1

A perspective: System 1 and System 2

System 1: thinking fast*, typically associated with* ***neural*** *models*
System 2: thinking slow*, typically associated with* ***symbolic*** *reasoning*

And for a long while the consensus was that we need neuro-symbolic solutions for difficult problems (i.e., problems that require System 2 type cognition)



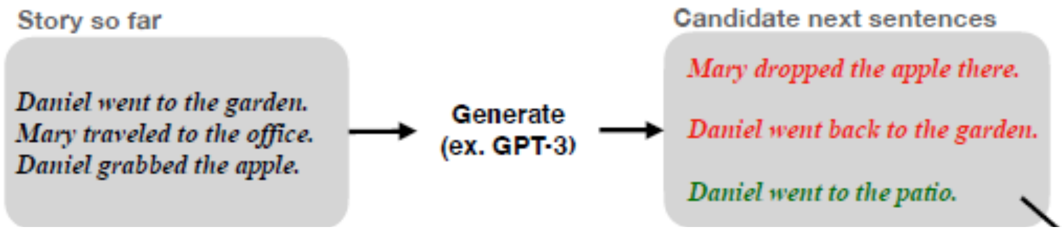
A Proof-of-Concept – System 1 + System 2

Improving Coherence and Consistency in Neural Sequence Models with Dual-System, Neuro-Symbolic Reasoning

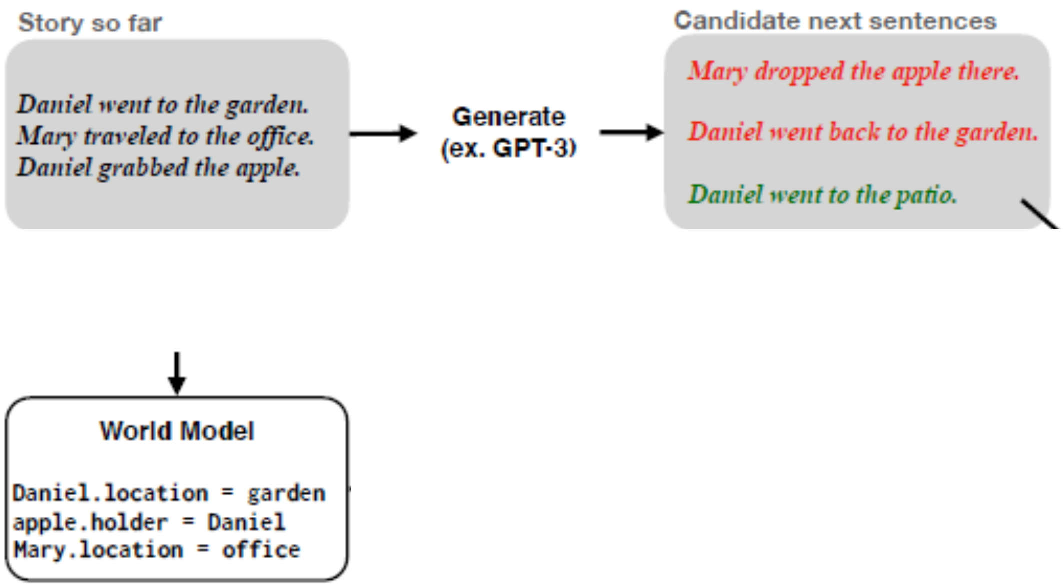
Maxwell Nye* — MIT
Michael Henry Tessler — MIT, DeepMind
Joshua B. Tenenbaum — MIT
Brenden M. Lake — NYU, Facebook AI Research

https://arxiv.org/pdf/2107.02794.pdf

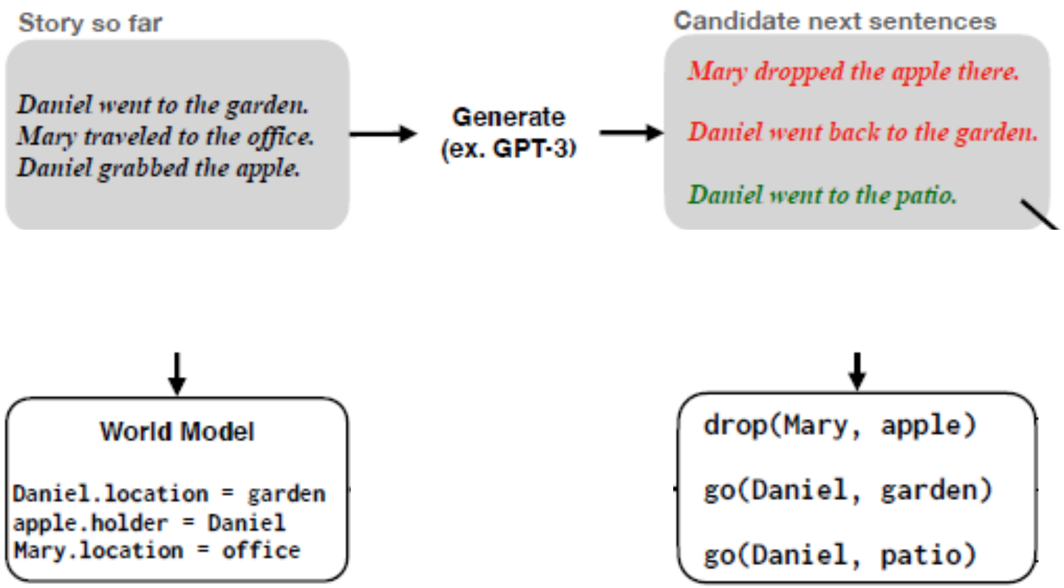
A Proof-of-Concept – GPT3



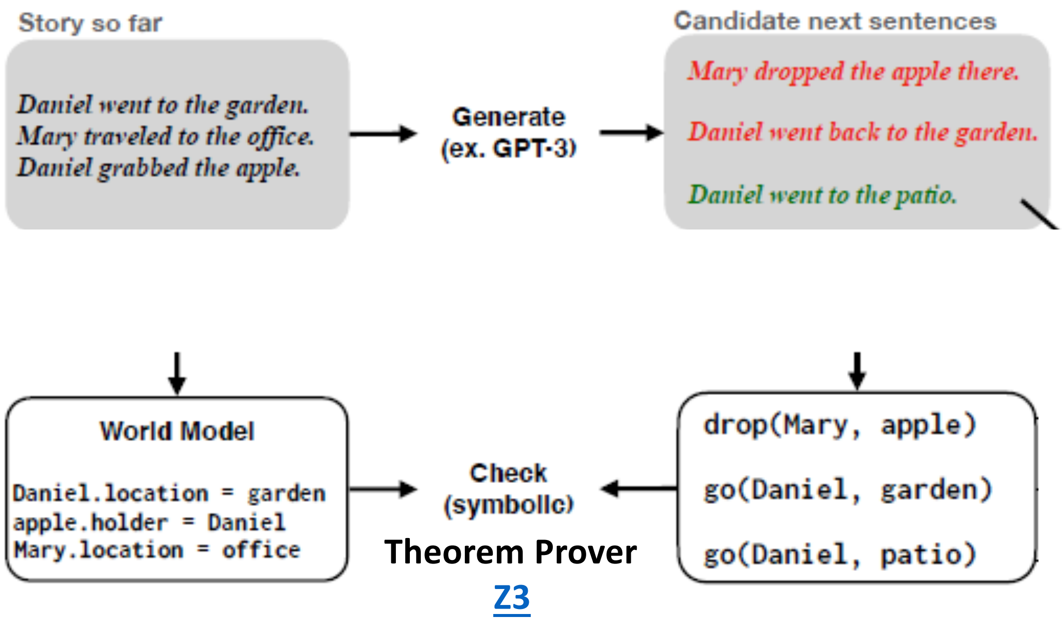
A Proof-of-Concept – GPT3



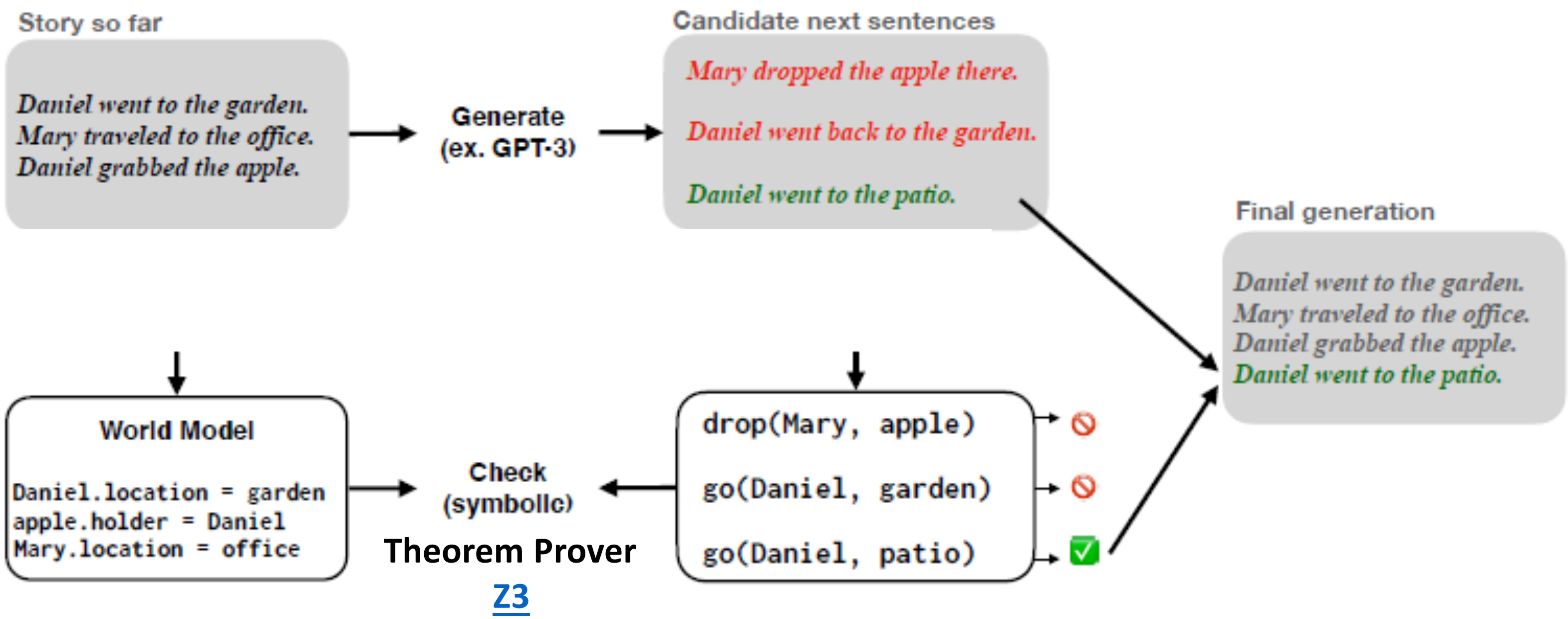
A Proof-of-Concept – GPT3



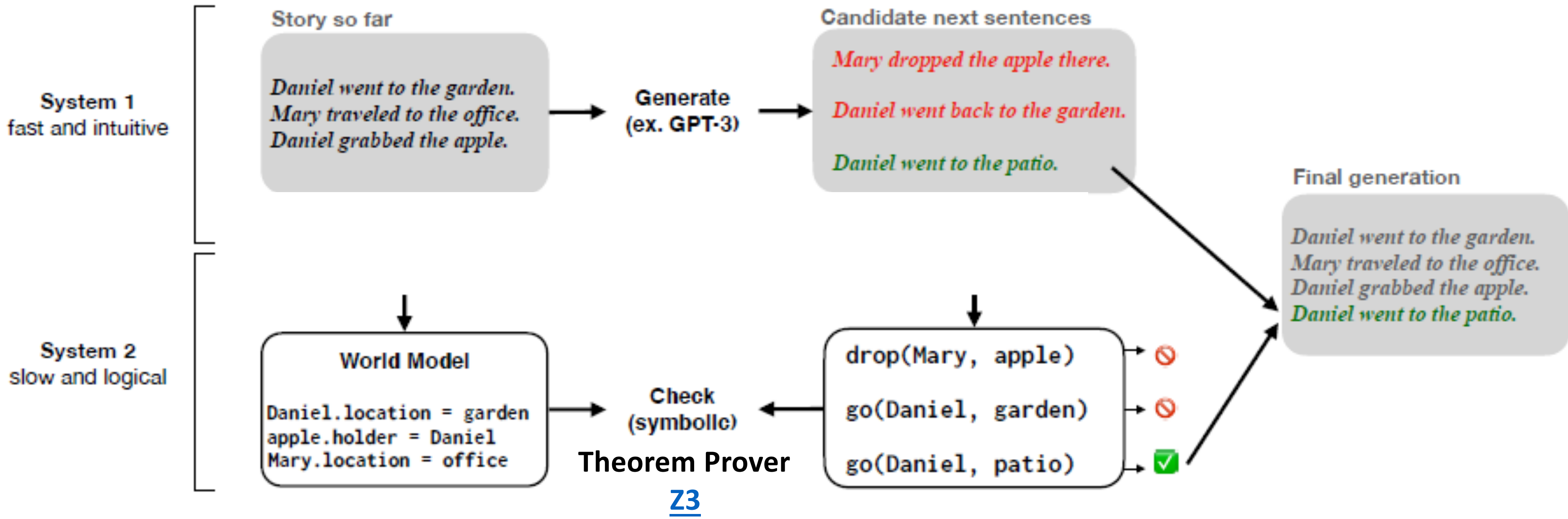
A Proof-of-Concept – GPT3



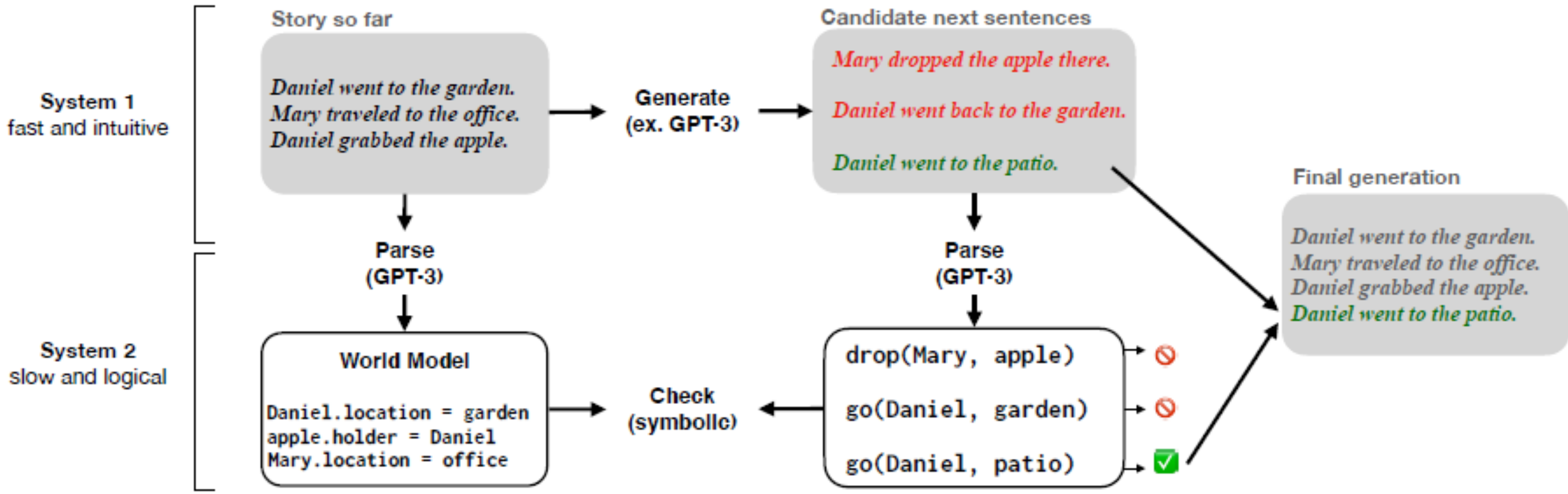
A Proof-of-Concept – GPT3



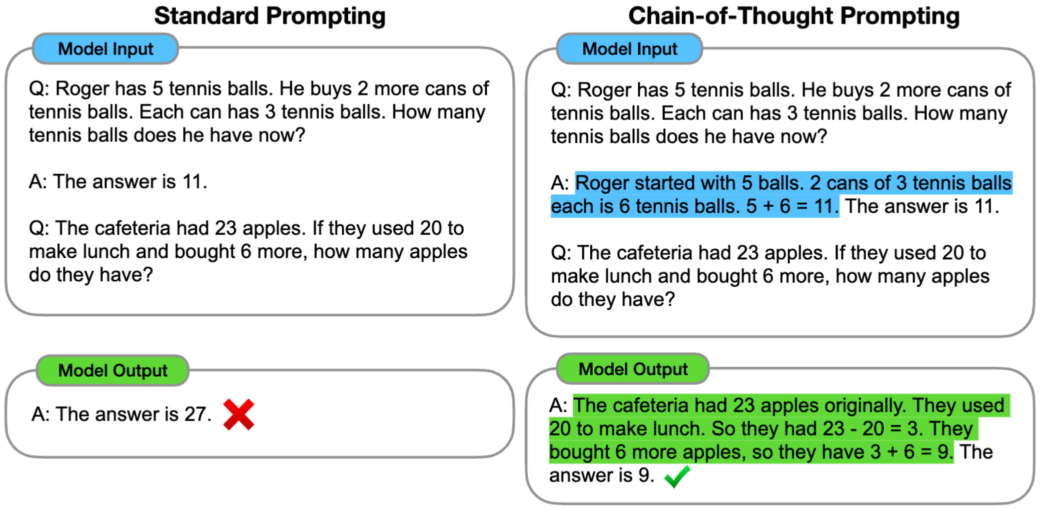
A Proof-of-Concept – GPT3

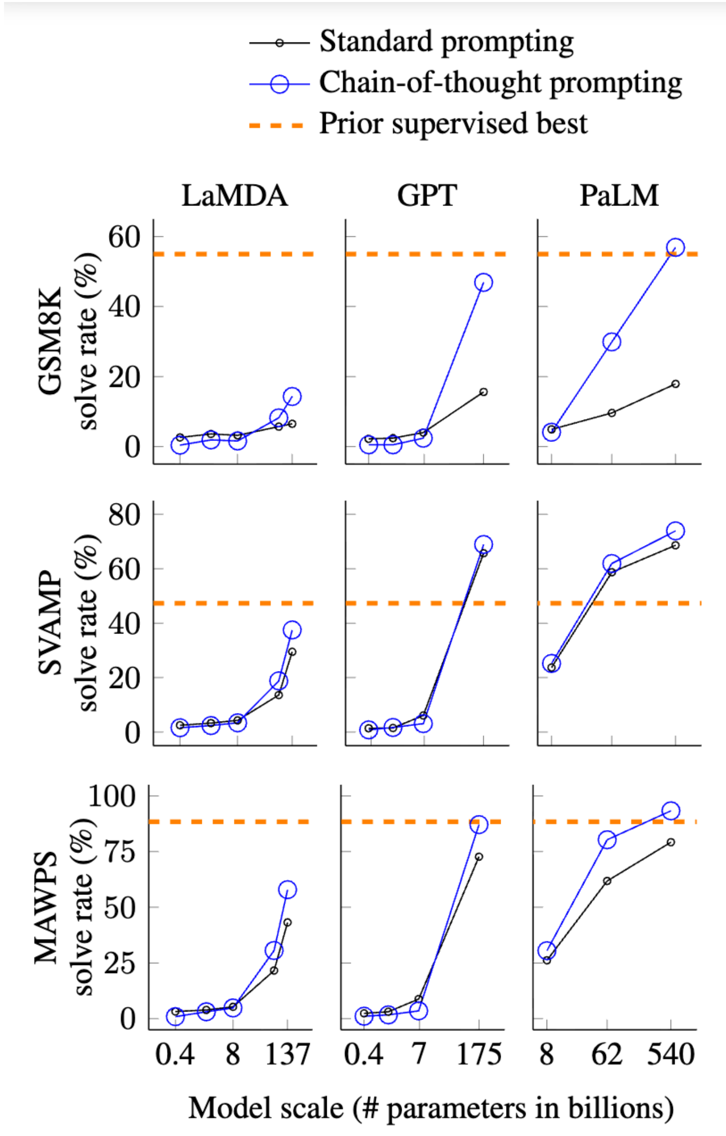


Teach GPT3 to produce symbolic outputs!



Chain of Thought (CoT) in LLMs





https://arxiv.org/pdf/2201.11903

Scratchpad

Show Your Work: Scratchpads for Intermediate Computation with Language Models

Maxwell Nye, Anders Johan Andreassen, Guy Gur-Ari, Henryk Michalewski, Jacob Austin, David Bieber, David Dohan, Aitor Lewkowycz, Maarten Bosma, David Luan, Charles Sutton, Augustus Odena

```
Input:
2 9 + 5 7

Target:
<scratch>
2 9 + 5 7 ,  C: 0
2 + 5 , 6 C: 1  # added 9 + 7 = 6 carry 1
, 8 6 C: 0  # added 2 + 5 + 1 = 8 carry 0
0 8 6
</scratch>
8 6
```

Scratchpad

Show Your Work: Scratchpads for Intermediate Computation with Language Models

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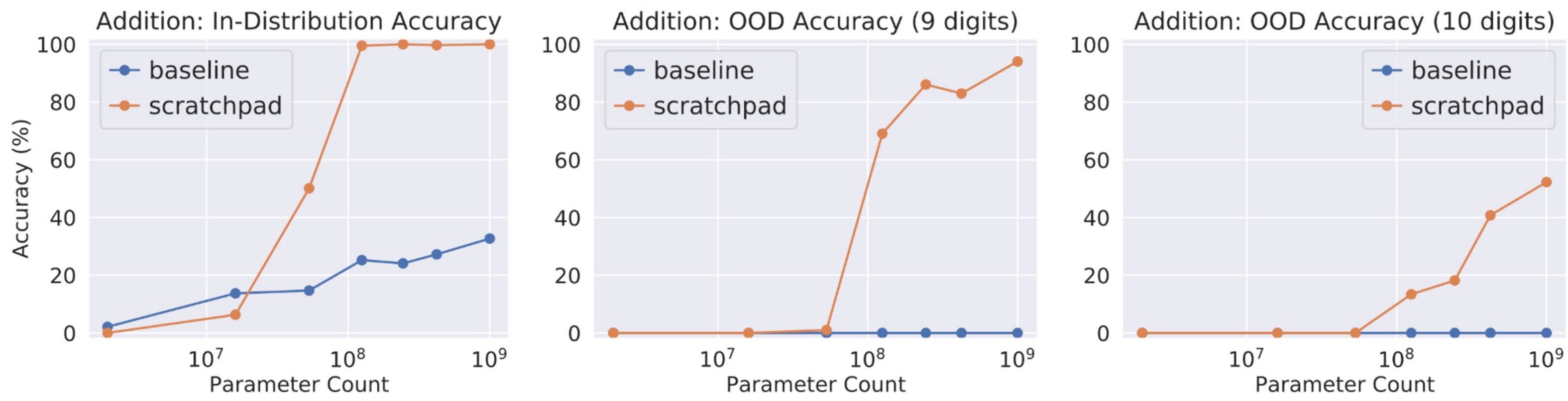


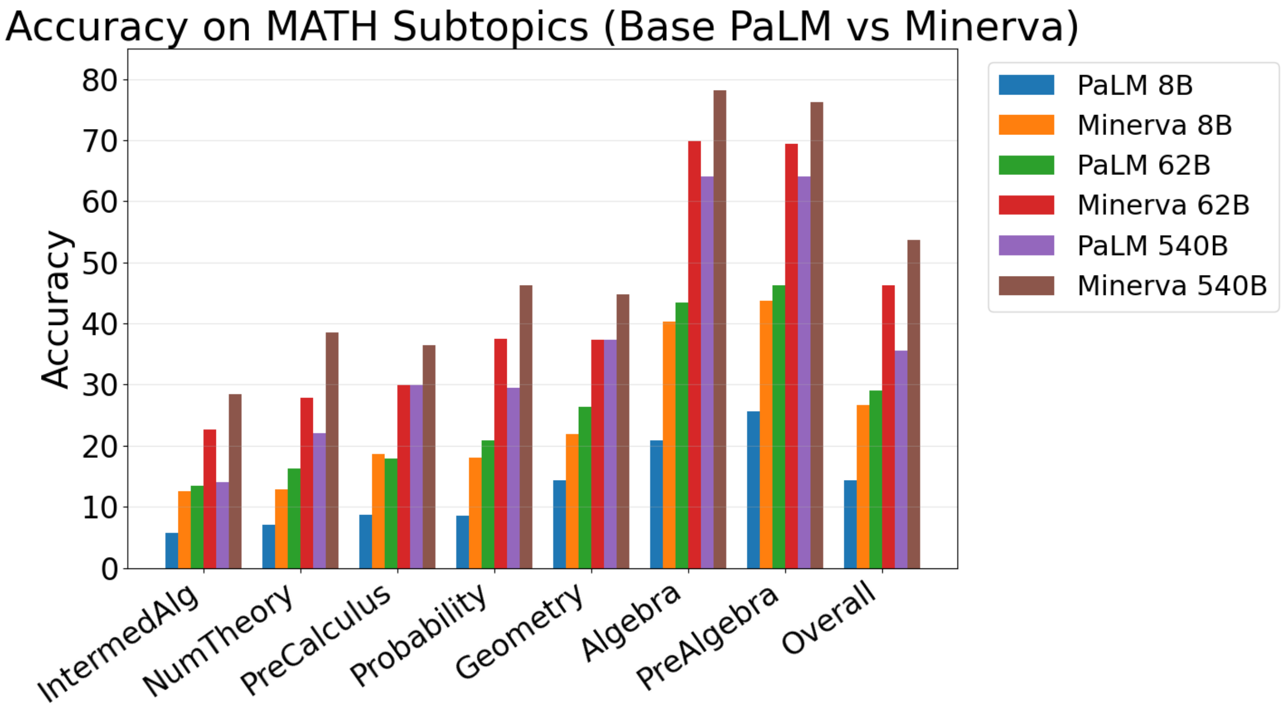
Figure 3: Using a scratchpad significantly improves the performance of pre-trained Transformer-based models on addition, including their ability to generalize out of the training distribution to numbers with more digits. Models were trained on 1-8 digit addition. The baseline models were trained without intermediate scratchpad steps.

Data is still important – Minerva model

builds on general-purpose pretrained models by *further training them on a large, high-quality dataset of scientific and mathematical technical content*.

Mathematical Web Pages (~17.5 B tokens)

arXiv Papers (~21 B tokens)



https://arxiv.org/abs/2206.14858

A simple model to explore CoT / reasoning

Assume the world is a joint distribution defined by a Bayes net.

The “brain” sees samples and learns a density over *sequences* of variables, q(“A=1,C=0,B=1”).

Estimate conditional distribution:

Directly: p(C=c|A=a) ~ q(c|“A=a,C=”)

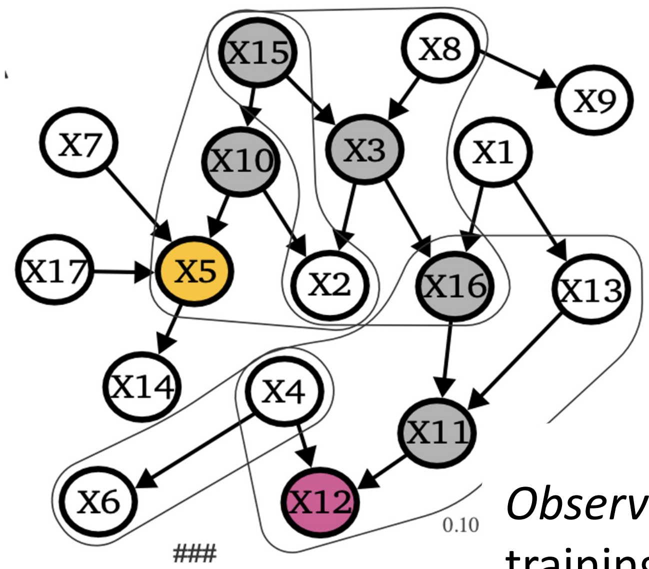
By sampling intermediate variables (a Monte Carlo estimate):
p(C=c|A=a) ~ q(c|”A=a,B=b,C=”) where “B=b,C=” ~ q

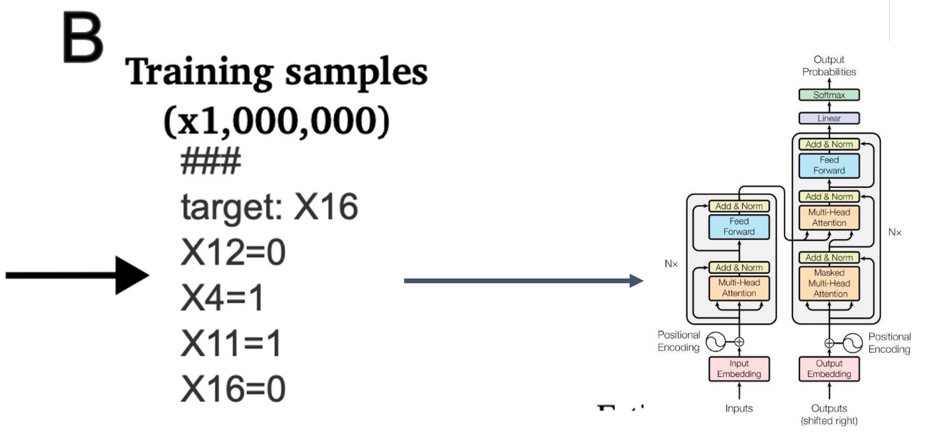
The *value of reasoning* is measured as the gap between *direct* and *free-generation* (i.e. Monte Carlo) estimators.

Prystawski, Li, Goodman (arxiv, 2023)

Slide from Noah Goodman

Training sequence density estimator (transformer)





Observation distribution samples subset of variables for each training sequence.

Compare *local* (Cf ego-centric) to *global* observation conditions.

Slide from Noah Goodman

Inference accuracy evaluation

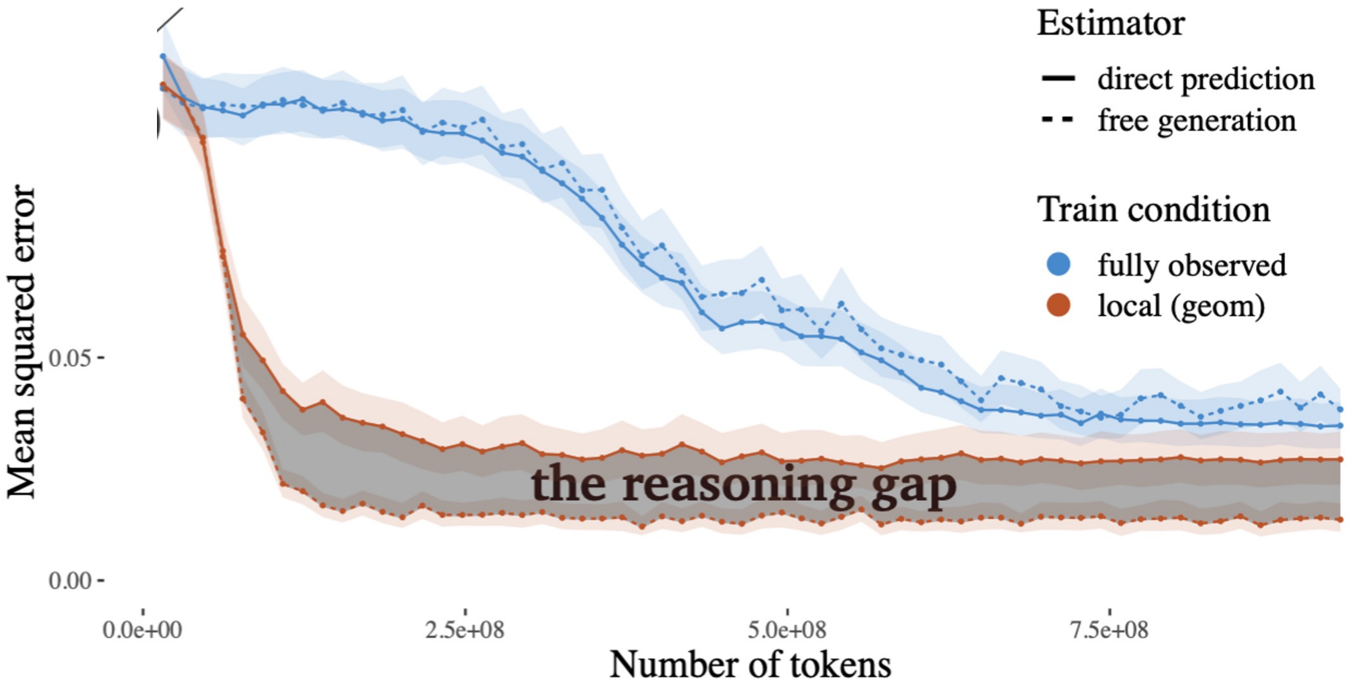
direct prediction

……
target: X5
X12=0
X5=_

free generation

###
target: X5
X12=0
X11=1
X16=0
X3=0
X15=1
X10=1
X5=_

model-generated



Slide from Noah Goodman

Results (summary)

When training data is organized into local observations (but not global or random) reasoning through intermediate variables ("Chain of Thought") is more accurate.

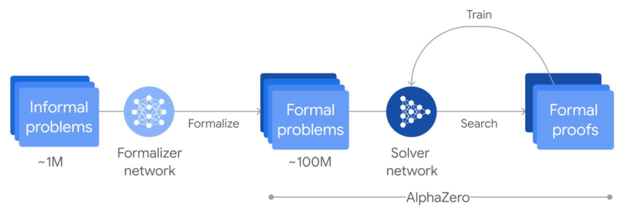
Theorem: For chain bayesnet and observations consisting of adjacent pairs of variables. A risk-minimizing sequence estimate q. The direct estimator has higher bias than the "Chain of Thought" estimator.

Data complexity is also vastly lower when using local observations.

So, reasoning (*viz a vis* CoT for sequence models) is useful when training data are organized locally.

July 25, 2024 Science

AI achieves silver-medal standard solving International Mathematical Olympiad problems



Process infographic of AlphaProof's reinforcement learning training loop: Around one million informal math problems are translated into a formal math language by a formalizer network. Then a solver network searches for proofs or disproofs of the problems, progressively training itself via the AlphaZero algorithm to solve more challenging problems.

AlphaProof: a formal approach to reasoning

AlphaProof is a system that trains itself to prove mathematical statements in the formal language Lean. It couples a pre-trained language model with the AlphaZero reinforcement learning algorithm, which previously taught itself how to master the games of chess, shogi and Go.

Formal languages offer the critical advantage that proofs involving mathematical reasoning can be formally verified for correctness. Their use in machine learning has, however, previously been constrained by the very limited amount of human-written data available.

In contrast, natural language based approaches can hallucinate plausible but incorrect intermediate reasoning steps and solutions, despite having access to orders of magnitudes more data. We established a bridge between these two complementary spheres by fine-tuning a Gemini model to automatically translate natural language problem statements into formal statements, creating a large library of formal problems of varying difficulty.

https://deepmind.google/blog/ai-solves-imo-problems-at-silver-medal-level/

July 21, 2025 Research

Advanced version of Gemini with Deep Think officially achieves gold-medal standard at the International Mathematical Olympiad

While our approach this year was based purely on natural language with Gemini, we also continue making progress on our formal systems, AlphaGeometry and AlphaProof. We believe agents that combine natural language fluency with rigorous reasoning - including verified reasoning in formal languages - will become invaluable tools for mathematicians, scientists, engineers, and researchers, helping us advance human knowledge on the path to AGI.

Andrew Lampinen
@AndrewLampinen

Our LM (and others) achieved gold-medal performance in the international Math Olympiad (x.com/demishassabis/...). Importantly, unlike last year's silver, this system used natural language end-to-end, rather than relying on a verifiable formal language.

> **Demis Hassabis** @demishassabis · Jul 21, 2025
> Official results are in - Gemini achieved gold-medal level in the International Mathematical Olympiad! 🏆 An advanced version was able to solve 5 out of 6 problems. Incredible progress - huge congrats to @lmthang and the team! deepmind.google/discover/blog/...

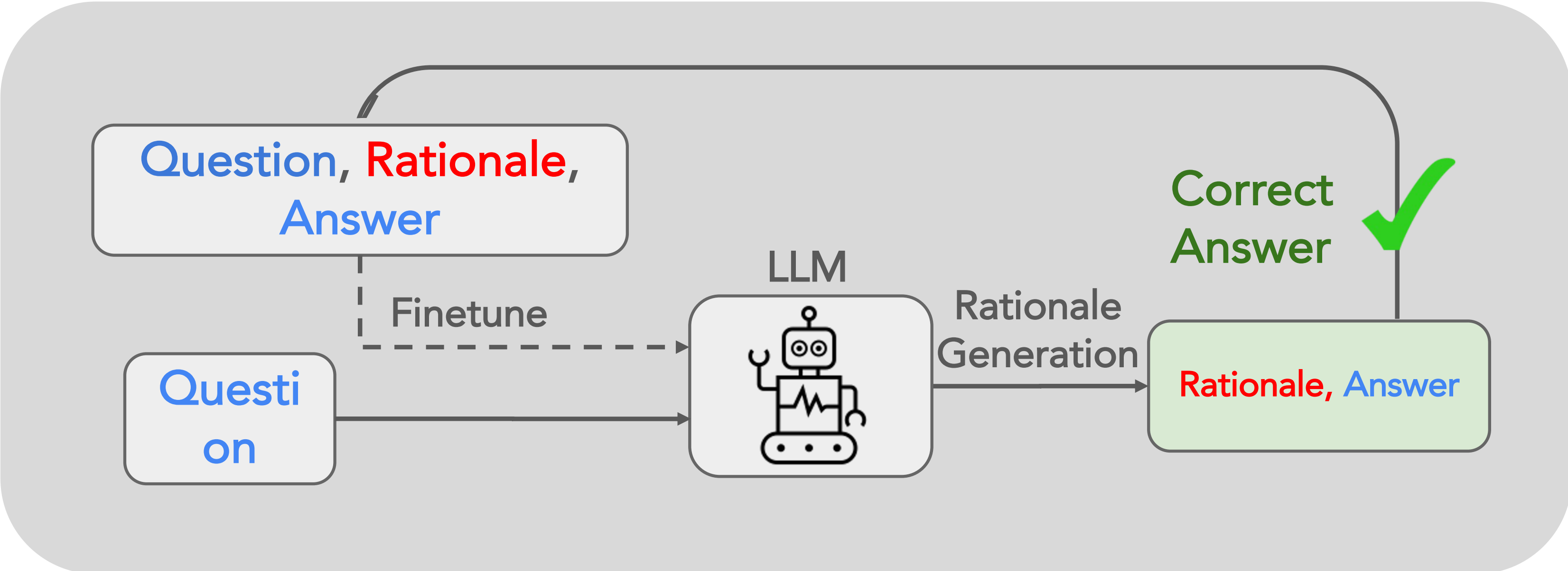
10:46 PM · Jul 21, 2025 · **11.9K** Views

https://deepmind.google/blog/advanced-version-of-gemini-with-deep-think-officially-achieves-gold-medal-standard-at-the-international-mathematical-olympiad/

System 2

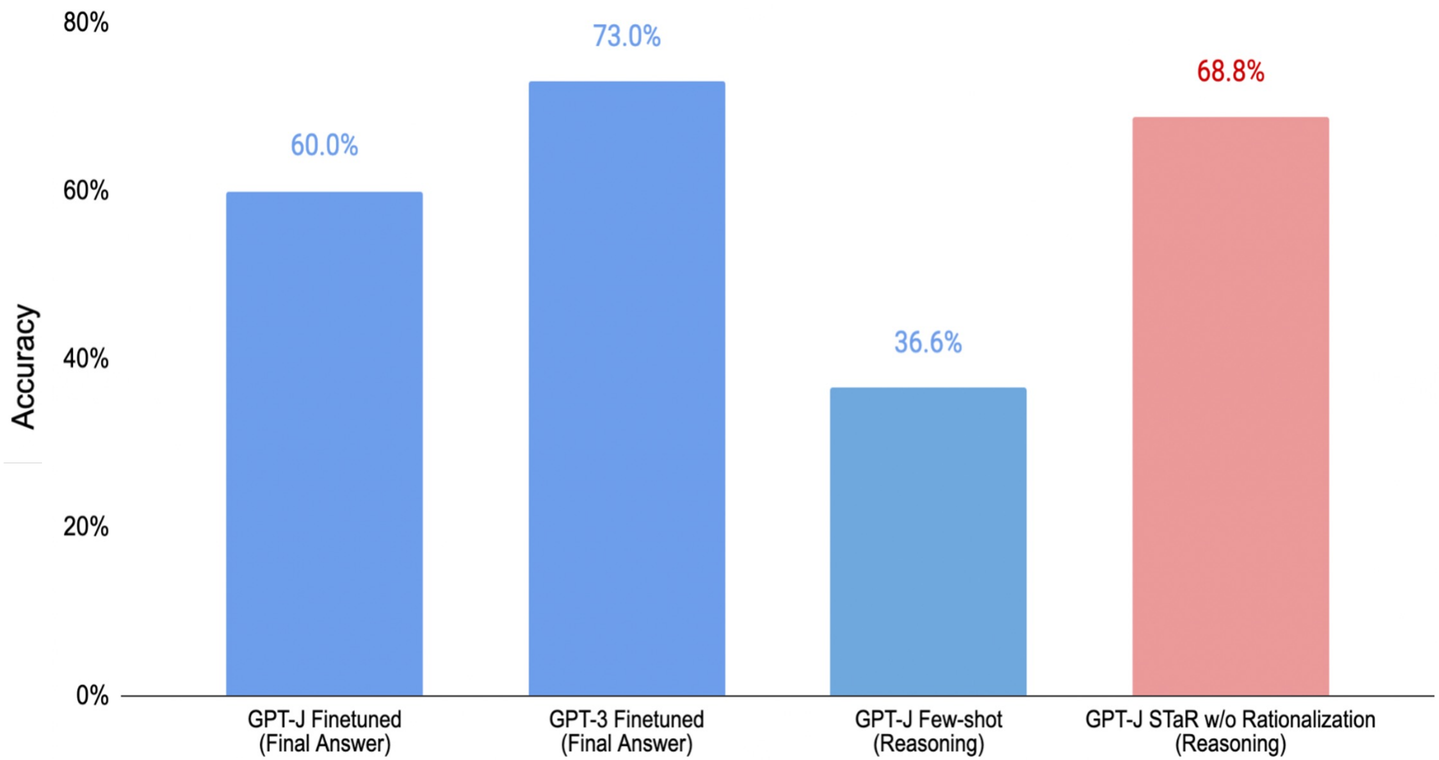
A talk by Yoshua Bengio on System 1 and 2: https://www.youtube.com/watch?v=FtUbMG3rlFs
And a “debate” between him and Gary Marcus: https://www.youtube.com/watch?v=EeqwFjqFvJA

Self-Taught Reasoner (STaR) - Basic version



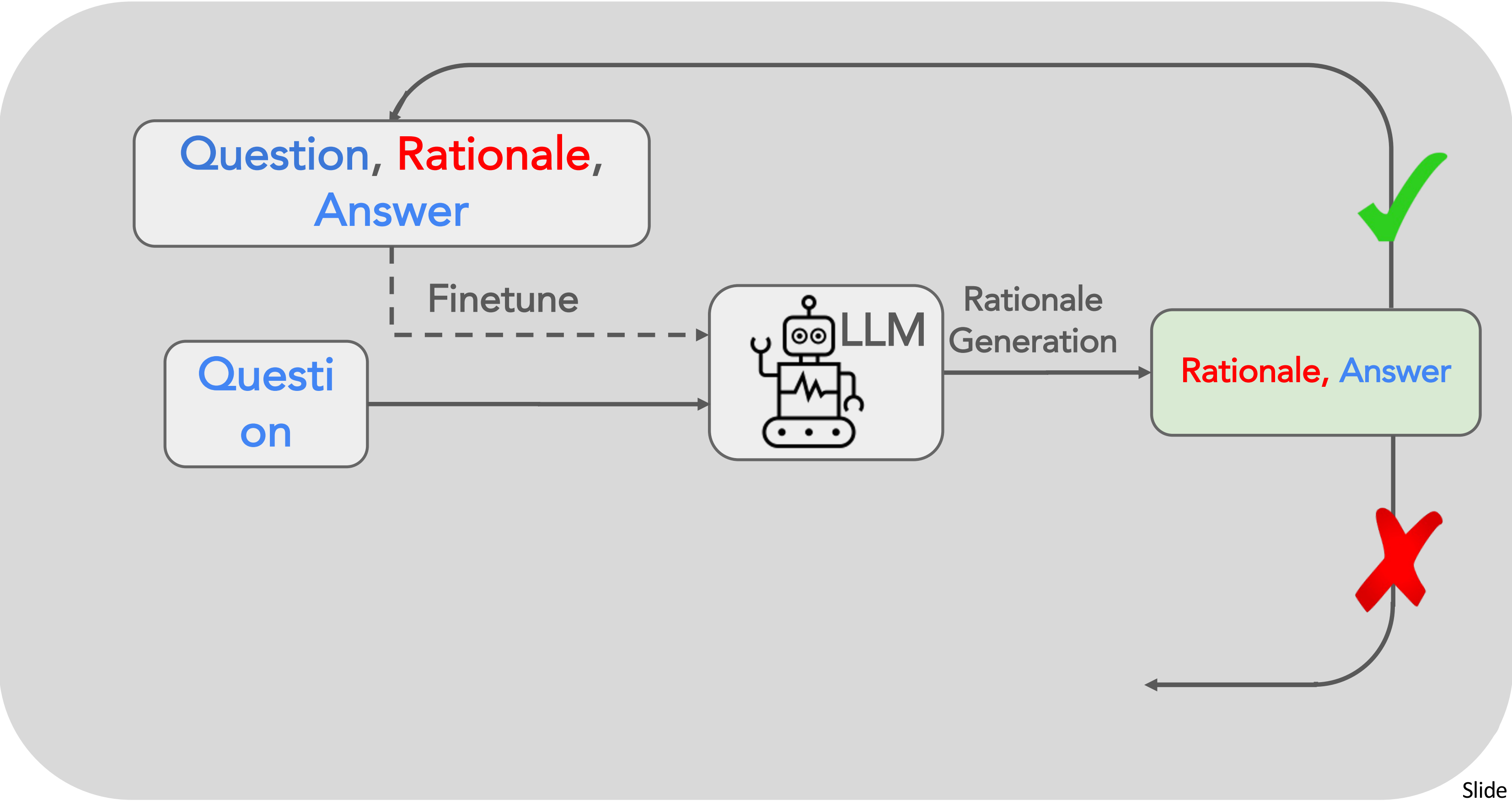
https://openreview.net/pdf?id=_3ELRdg2sgI

CommonsenseQA with STaR (Basic)



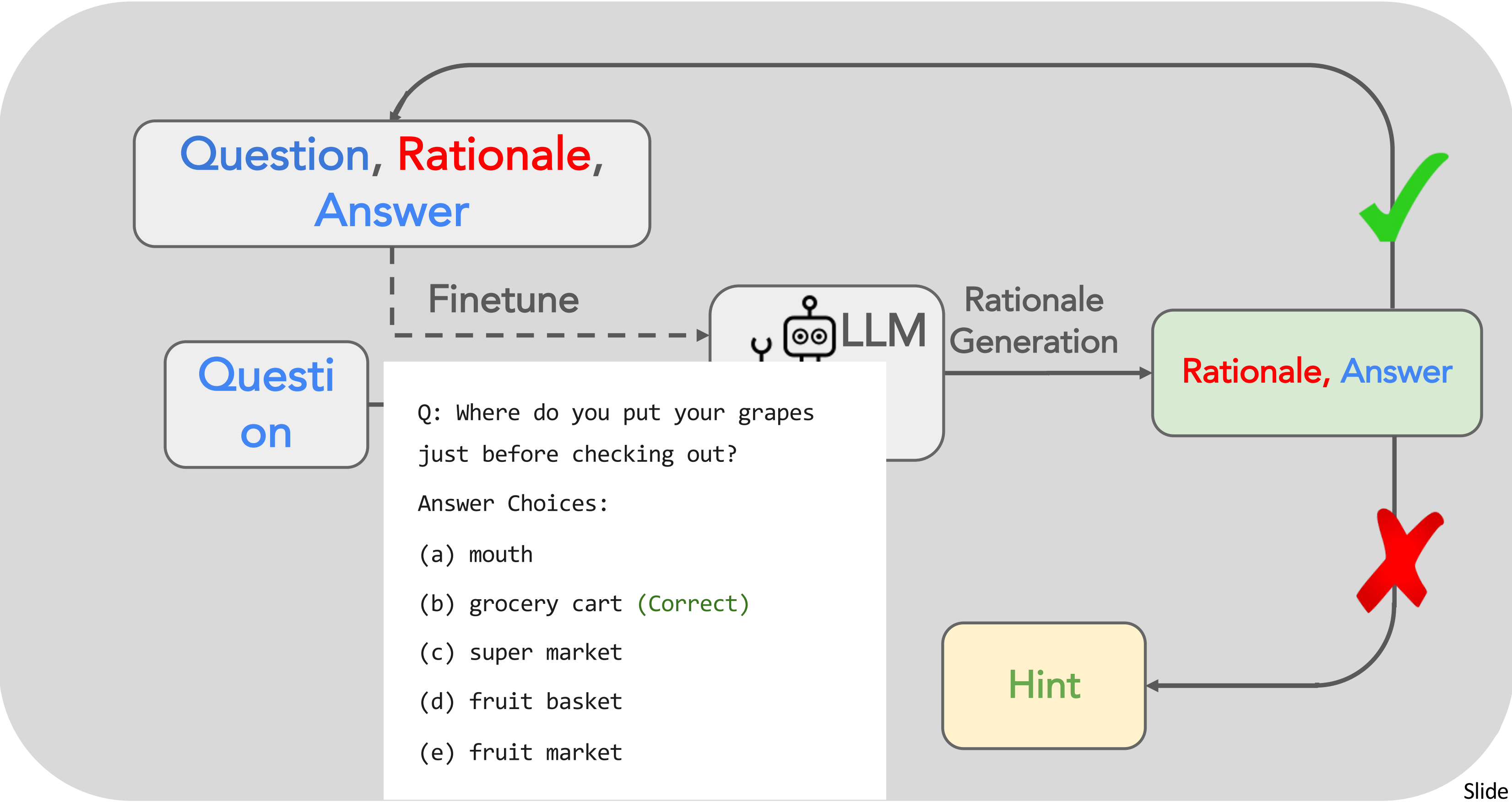
Slide from Noah Goodman

What to do with the wrong answers?



Slide from Noah Goodman

What to do with the wrong answers?



Q: Where do you put your grapes just before checking out?

Answer Choices:

(a) mouth

(b) grocery cart (Correct)

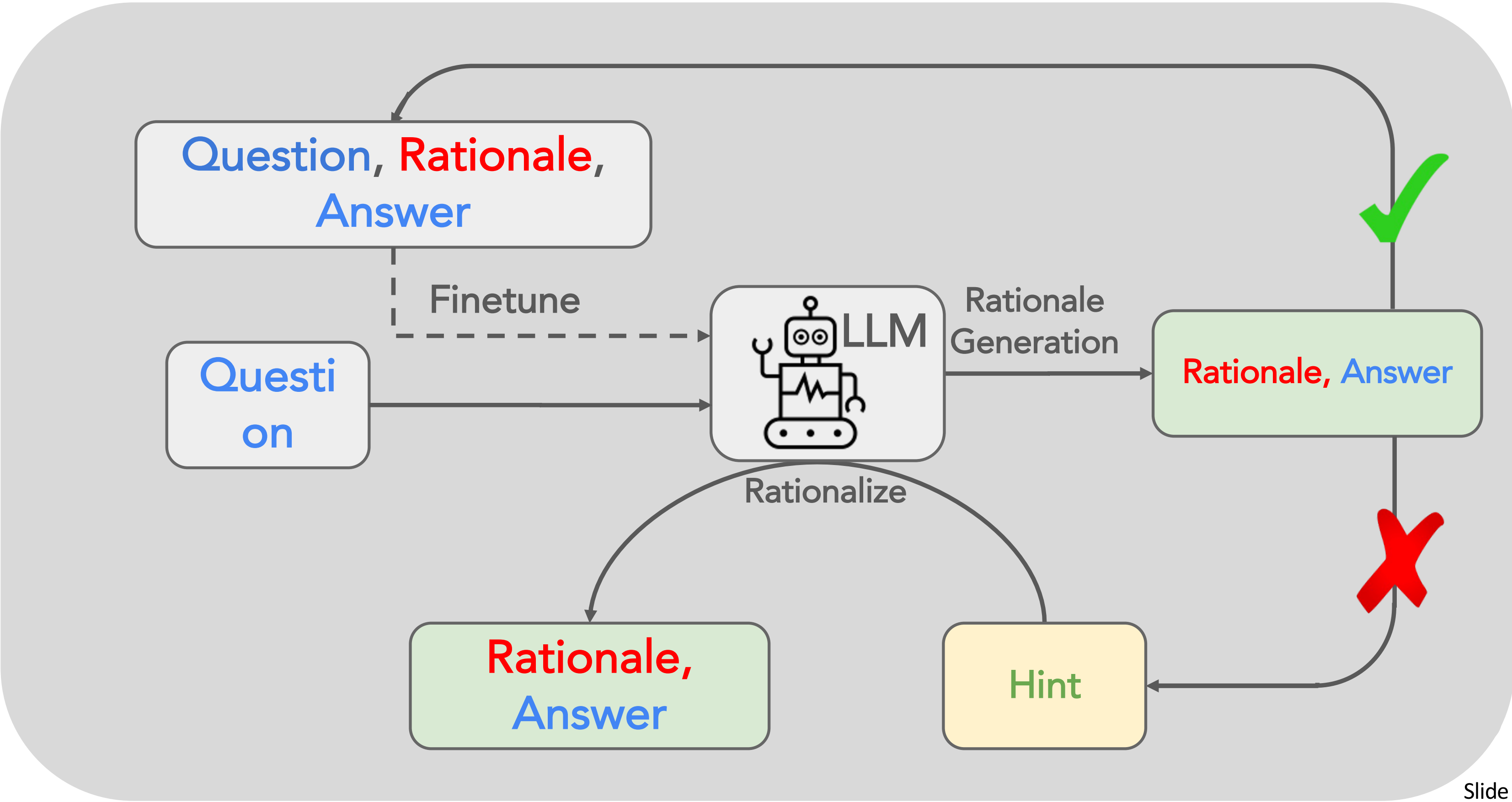
(c) super market

(d) fruit basket

(e) fruit market

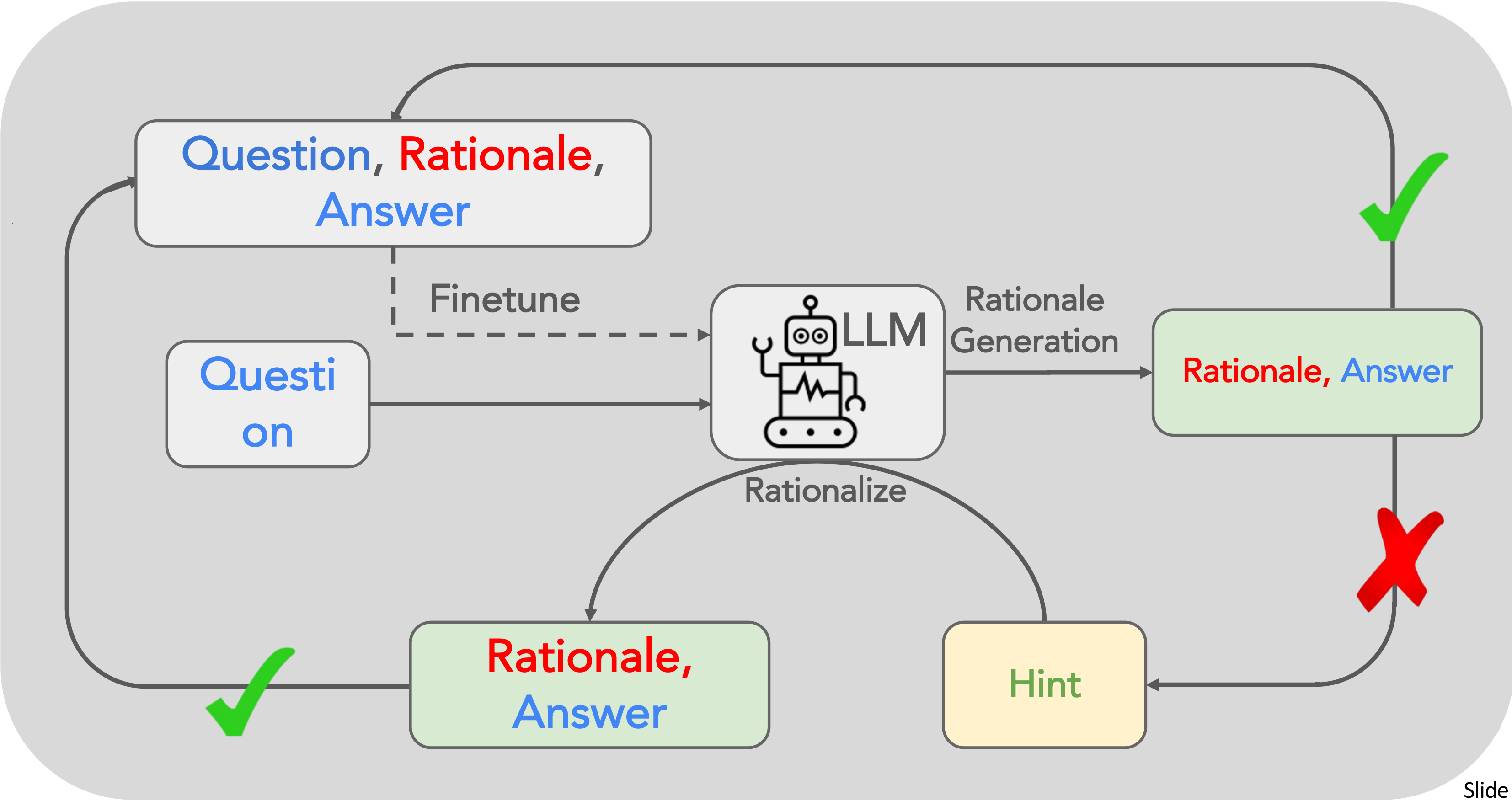
Slide from Noah Goodman

What to do with the wrong answers?



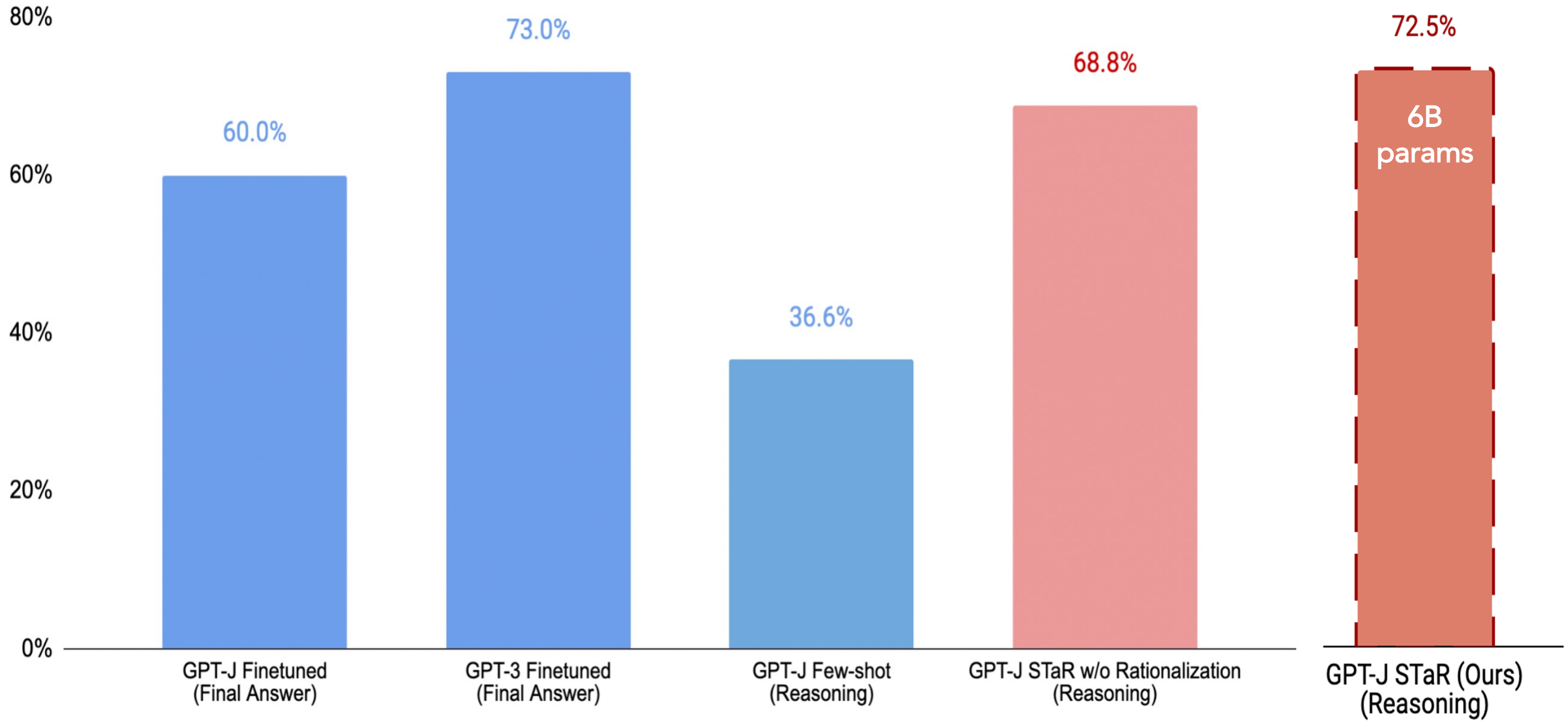
Slide from Noah Goodman

What to do with the wrong answers?



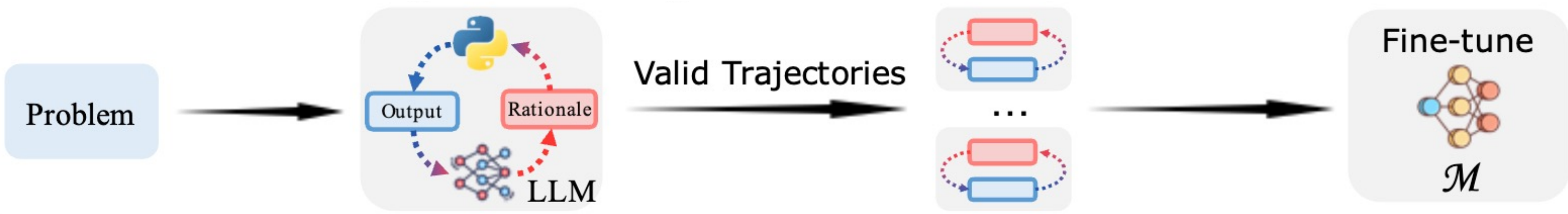
Slide from Noah Goodman

CommonsenseQA with STaR



Slide from Noah Goodman

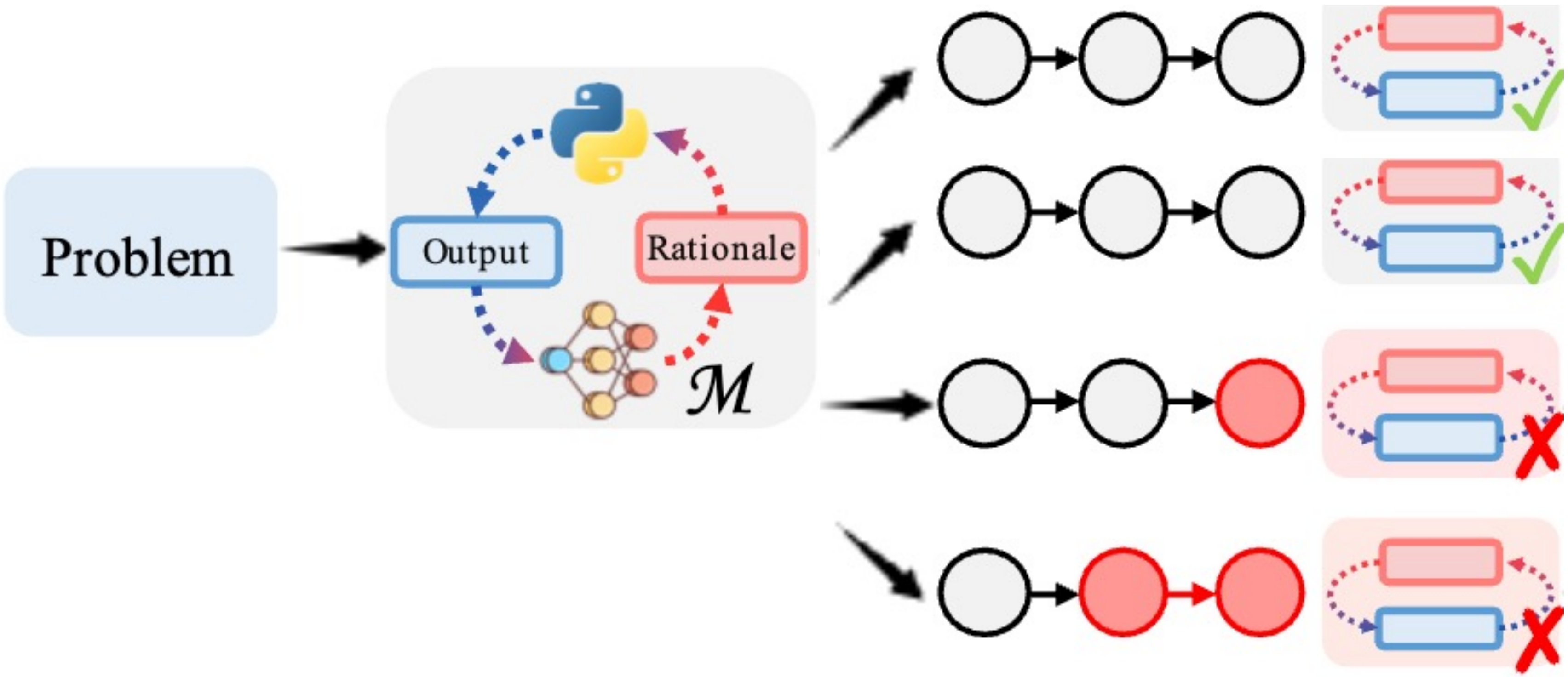
Imitation Learning works too



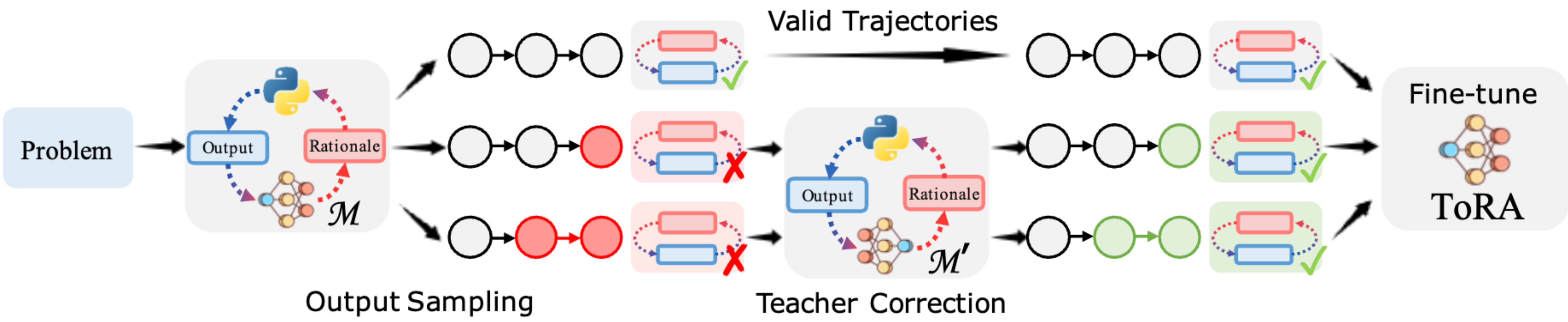
Prompt LLMs like GPT-4 to generate reasoning trajectories and use this collected data to fine-tune another, smaller, or weaker model

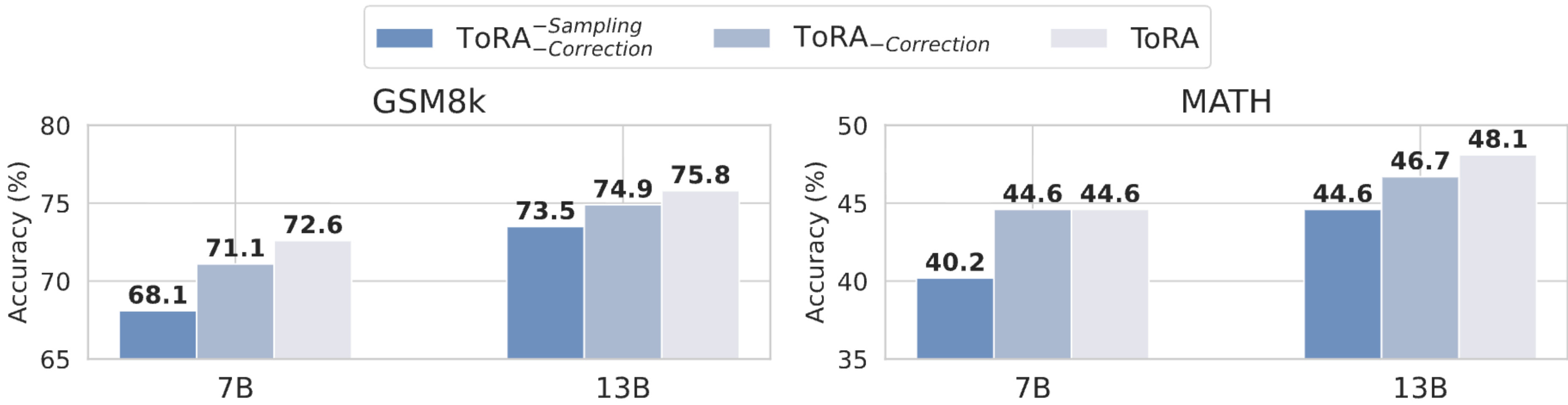
https://arxiv.org/pdf/2309.17452

Collecting diverse solutions for the same problems



Fixing failed trajectories using teacher model




ToRA$^{-Sampling}_{-Correction}$
ToRA$_{-Correction}$
ToRA
GSM8k
MATH
Accuracy (%)
80
75
70
65
50
45
40
35
68.1
71.1
72.6
73.5
74.9
75.8
40.2
44.6
44.6
44.6
46.7
48.1
7B
13B
7B
13B

Using RL to teach model to reason:
The Deepseek R1

Deepseek R1 (Reasoning LLMs)

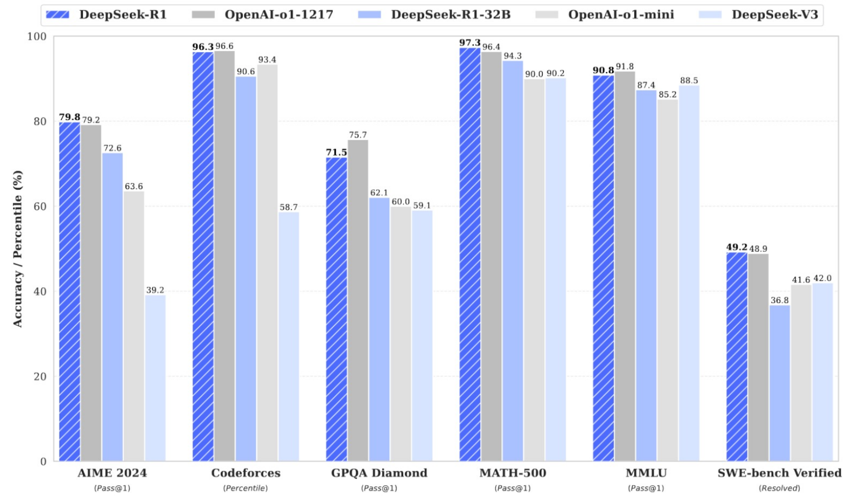
DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI

research@deepseek.com

Abstract

We introduce our first-generation reasoning models, DeepSeek-R1-Zero and DeepSeek-R1. DeepSeek-R1-Zero, a model trained via large-scale reinforcement learning (RL) without supervised fine-tuning (SFT) as a preliminary step, demonstrates remarkable reasoning capabilities. Through RL, DeepSeek-R1-Zero naturally emerges with numerous powerful and intriguing reasoning behaviors. However, it encounters challenges such as poor readability, and language mixing. To address these issues and further enhance reasoning performance, we introduce DeepSeek-R1, which incorporates multi-stage training and cold-start data before RL. DeepSeek-R1 achieves performance comparable to OpenAI-o1-1217 on reasoning tasks. To support the research community, we open-source DeepSeek-R1-Zero, DeepSeek-R1, and six dense models (1.5B, 7B, 8B, 14B, 32B, 70B) distilled from DeepSeek-R1 based on Qwen and Llama.



Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
OpenAI-o1-0912	74.4	83.3	94.8	77.3	63.4	1843
DeepSeek-R1-Zero	71.0	86.7	95.9	73.3	50.0	1444

Table 2 | Comparison of DeepSeek-R1-Zero and OpenAI o1 models on reasoning-related benchmarks.

https://arxiv.org/pdf/2501.12948

Group Relative Policy Optimization (GRPO)

Specifically, for each question q, GRPO samples a group of outputs $\{o_1, o_2, \cdots, o_G\}$ from the old policy $\pi_{\theta_{old}}$ and then optimizes the policy model π_θ by maximizing the following objective:

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{old}}(O|q)]$$

$$\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}A_i, \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon\right)A_i\right) - \beta\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right)\right), \tag{1}$$

$$\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right) = \frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - 1, \tag{2}$$

where ε and β are hyper-parameters, and A_i is the advantage, computed using a group of rewards $\{r_1, r_2, \ldots, r_G\}$ corresponding to the outputs within each group:

$$A_i = \frac{r_i - mean(\{r_1, r_2, \cdots, r_G\})}{std(\{r_1, r_2, \cdots, r_G\})}. \tag{3}$$

https://huggingface.co/docs/trl/main/en/grpo_trainer

What is the reward in Deepseek-R1-Zero? Bunch of rules

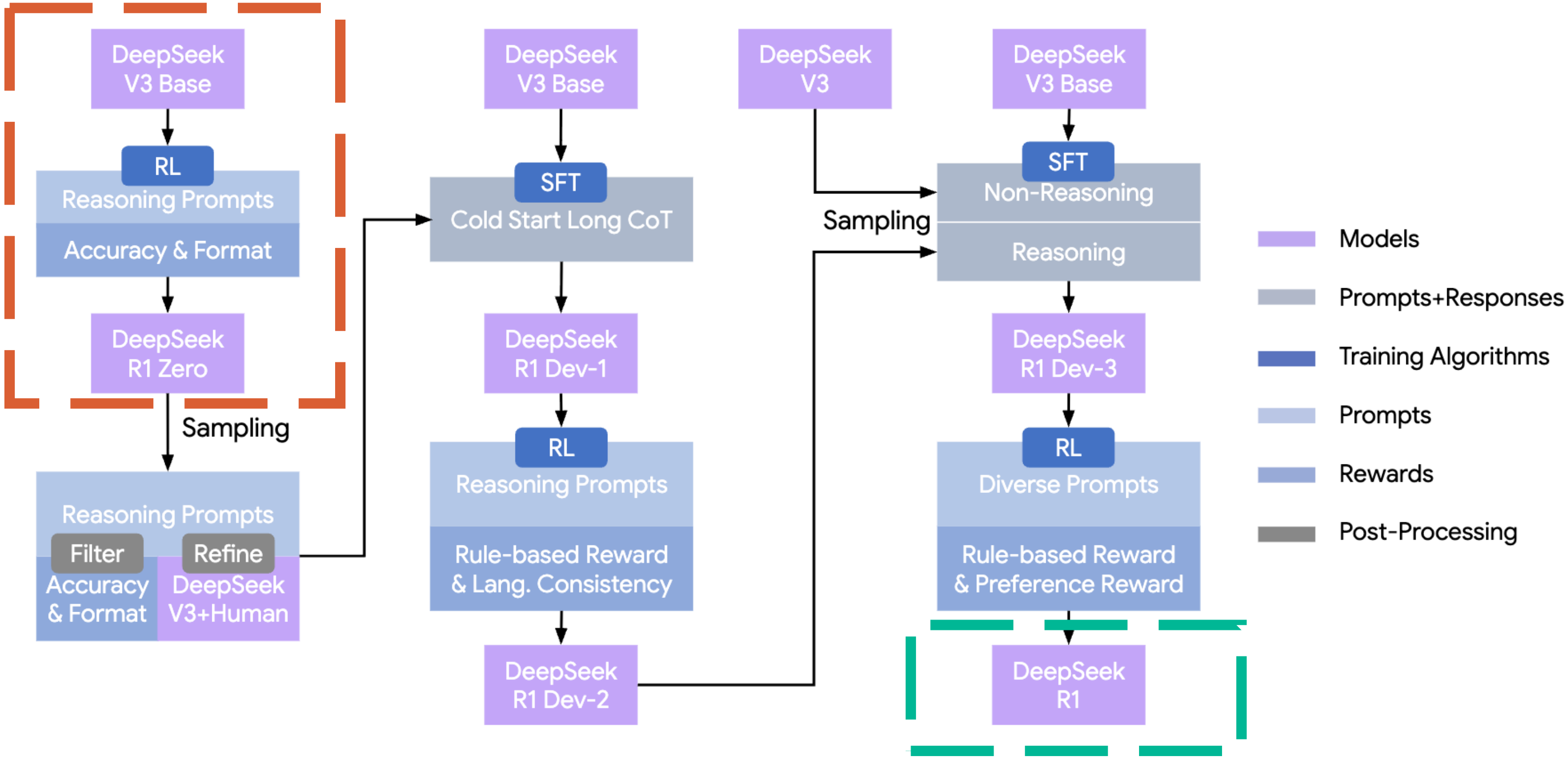
The reward is the source of the training signal, which decides the optimization direction of RL. To train DeepSeek-R1-Zero, we adopt a rule-based reward system that mainly consists of two types of rewards:

- **Accuracy rewards**: The accuracy reward model evaluates whether the response is correct. For example, in the case of math problems with deterministic results, the model is required to provide the final answer in a specified format (e.g., within a box), enabling reliable rule-based verification of correctness. Similarly, for LeetCode problems, a compiler can be used to generate feedback based on predefined test cases.
- **Format rewards**: In addition to the accuracy reward model, we employ a format reward model that enforces the model to put its thinking process between '<think>' and '</think>' tags.

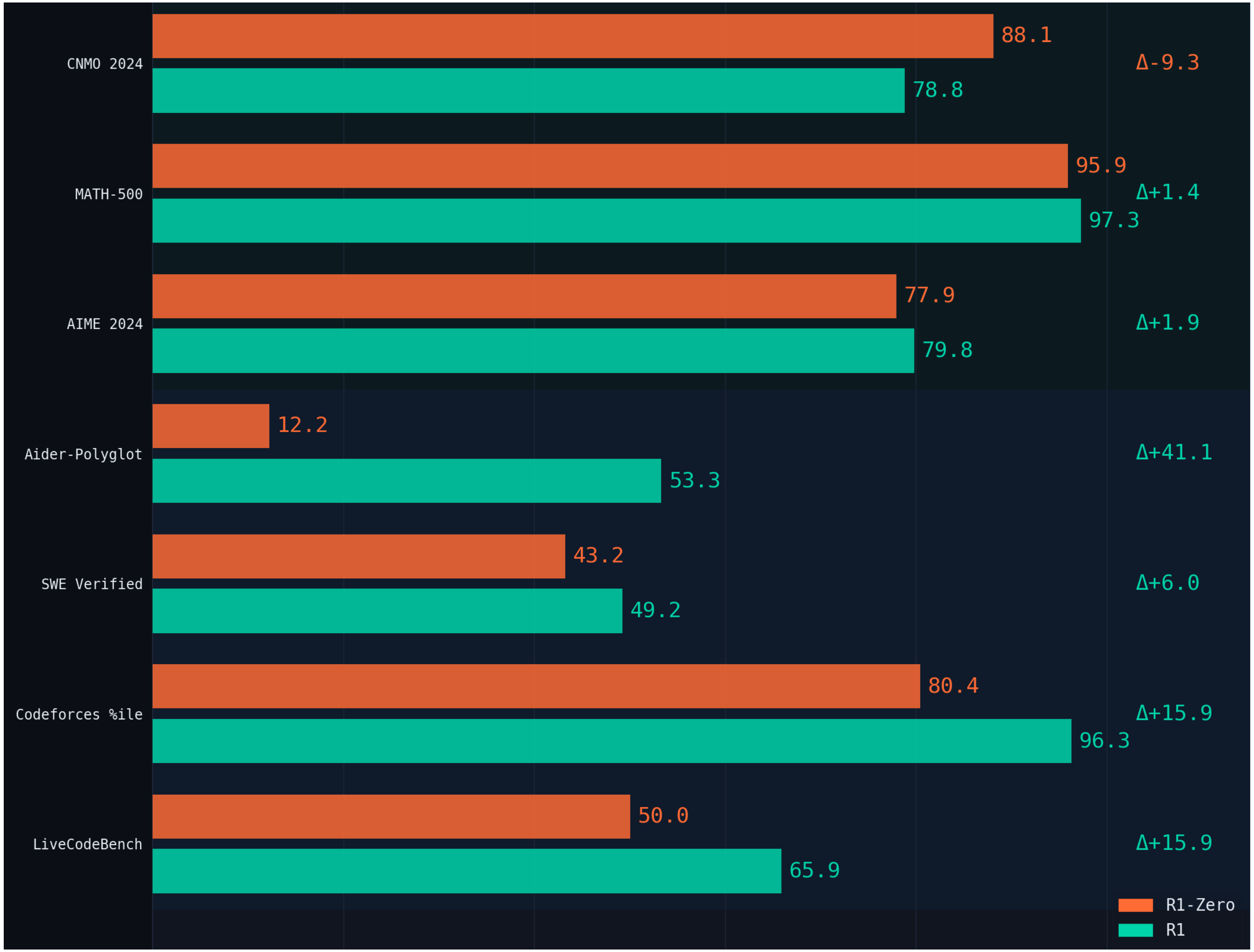
A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within <think> </think> and <answer> </answer> tags, respectively, i.e., <think> reasoning process here </think> <answer> answer here </answer>. User: prompt. Assistant:

Table 1 | Template for DeepSeek-R1-Zero. prompt will be replaced with the specific reasoning question during training.

The multi-stage pipeline of DeepSeek-R1



The basic Zero-R1 is competitive



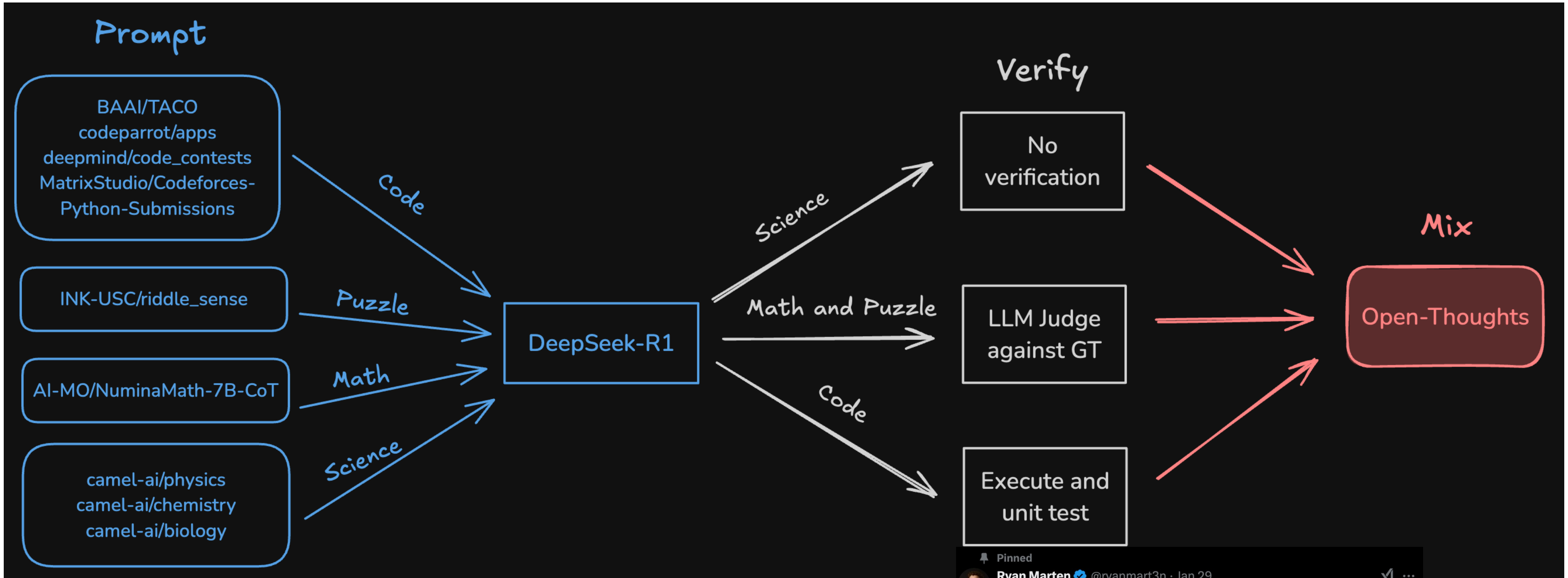
"The self-evolution of DeepSeek-R1-Zero exemplifies how RL can autonomously enhance a model's reasoning capabilities."

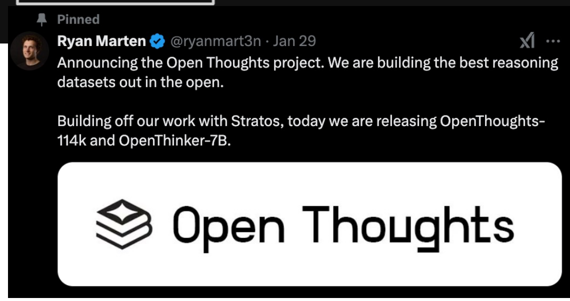
Smaller LMs: RL from scratch vs. Distillation

Model	AIME 2024		MATH-500
	pass@1	cons@64	pass@1
QwQ-32B-Preview	50.0	60.0	90.6
DeepSeek-R1-Zero-Qwen-32B	47.0	60.0	91.6
DeepSeek-R1-Distill-Qwen-32B	**72.6**	**83.3**	**94.3**

DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9
DeepSeek-R1-Distill-Qwen-32B	**72.6**	83.3	94.3
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1
DeepSeek-R1-Distill-Llama-70B	70.0	**86.7**	**94.5**

Distillation from Stronger Reasoners

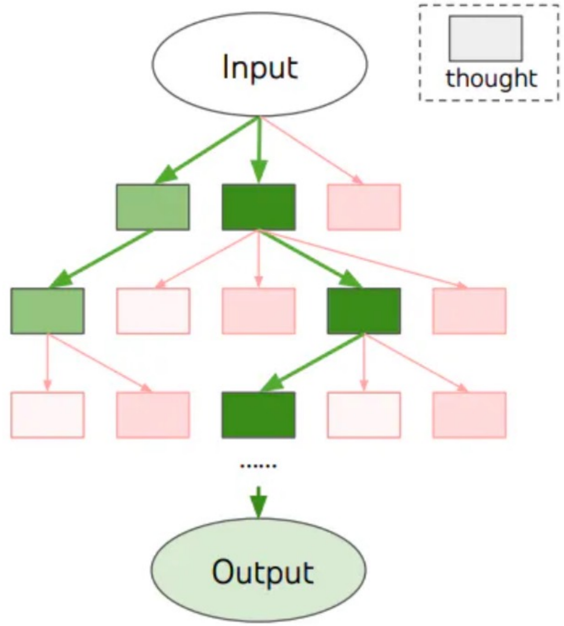




https://github.com/open-thoughts/open-thoughts

Reminder: Tree-of-thought

Or searching <u>externally</u>

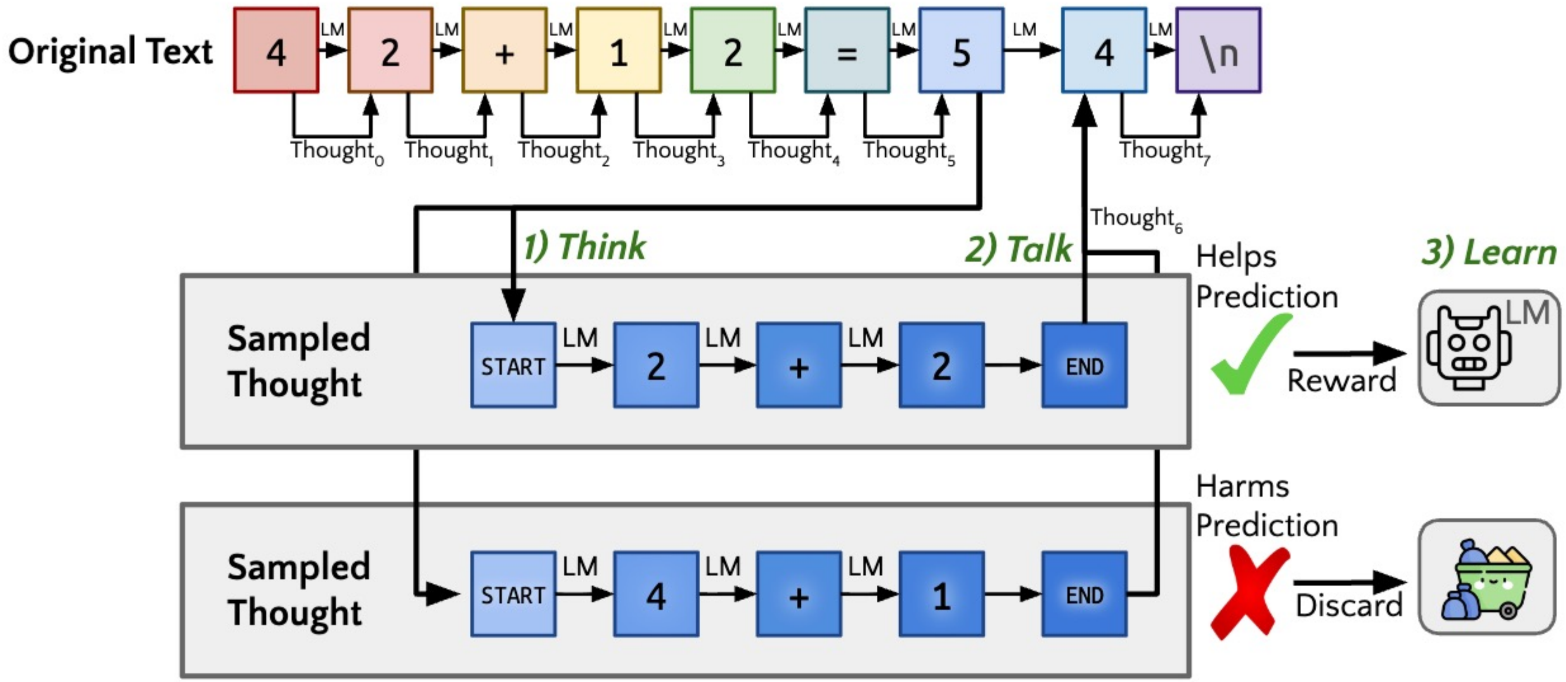


https://arxiv.org/pdf/2305.10601.pdf

Thinking before generating

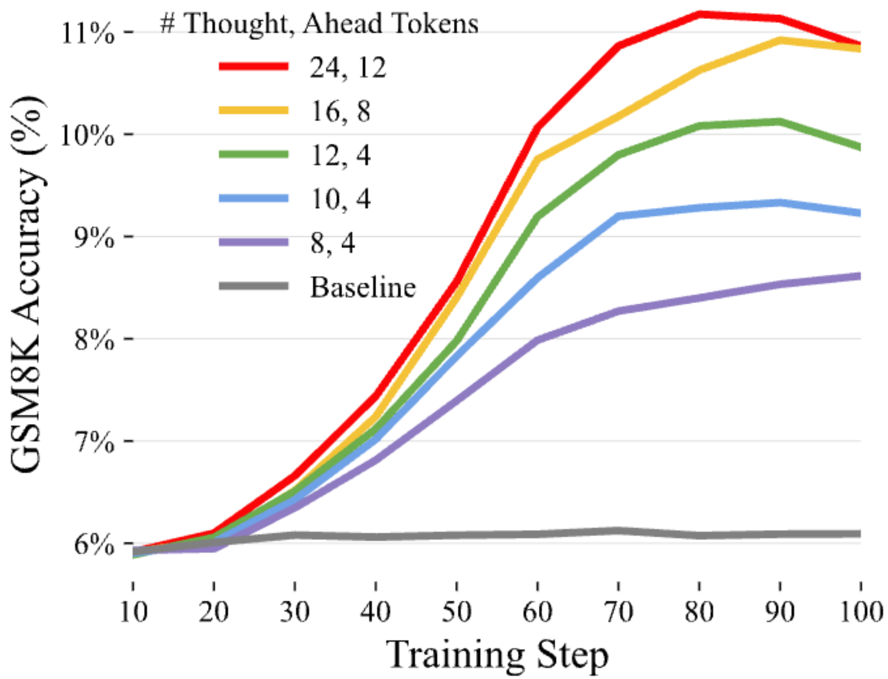
Or searching internally

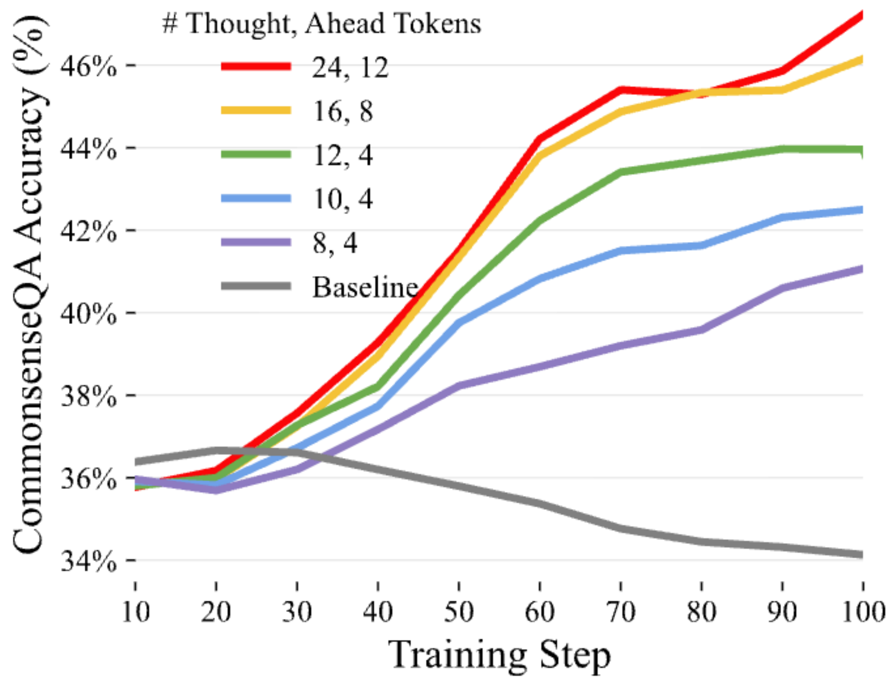
Think Before Speaking



https://arxiv.org/pdf/2403.09629

Think Before Speaking





https://arxiv.org/pdf/2403.09629

Inference Time Scaling

The O1/O2/etc

We saw training …

Now two shifts in inference paradigm

Inference Time Scaling

Generate many plausible responses by default

Verification (will cover these in the next lecture)

Verify those responses and return the best one

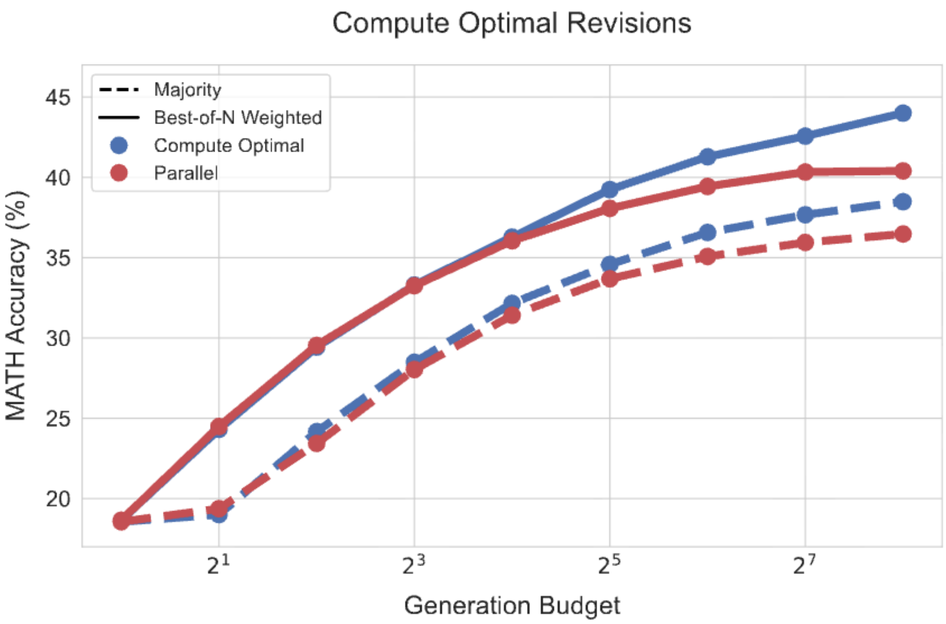
Scaling Test-Time Compute

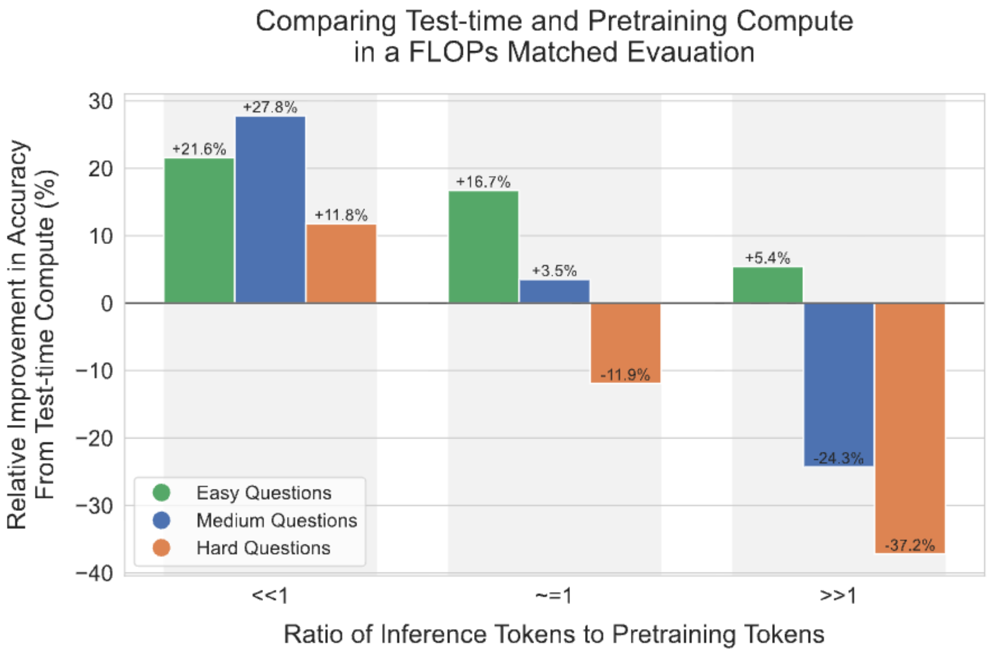
Google DeepMind 2024-8-7

Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters

Charlie Snell[♣,1], Jaehoon Lee[2], Kelvin Xu[♣,2] and Aviral Kumar[♣,2]

[♣]Equal advising, [1]UC Berkeley, [2]Google DeepMind, [♦]Work done during an internship at Google DeepMind





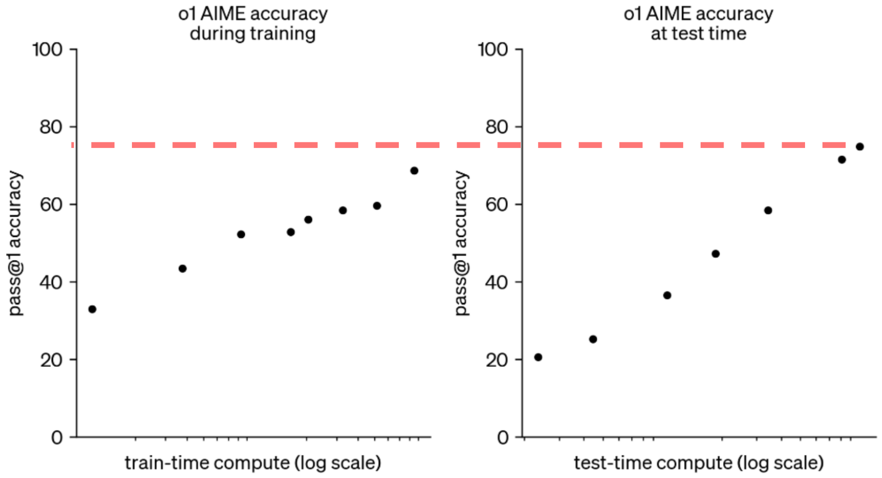
https://arxiv.org/pdf/2408.03314

Inference Time Scaling (emergence of OpenAI O1)

September 12, 2024 Release

Learning to reason with LLMs

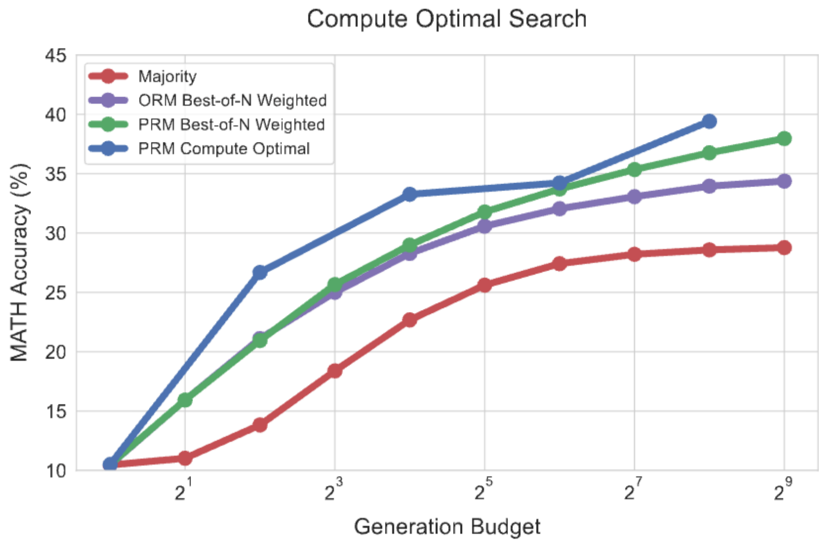
We are introducing OpenAI o1, a new large language model trained with reinforcement learning to perform complex reasoning. o1 thinks before it answers—it can produce a long internal chain of thought before responding to the user.



https://openai.com/index/learning-to-reason-with-llms/

Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters

Charlie Snell[♦,1], Jaehoon Lee[2], Kelvin Xu[♣,2] and Aviral Kumar[♣,2]
[♣]Equal advising, [1]UC Berkeley, [2]Google DeepMind, [♦]Work done during an internship at Google DeepMind

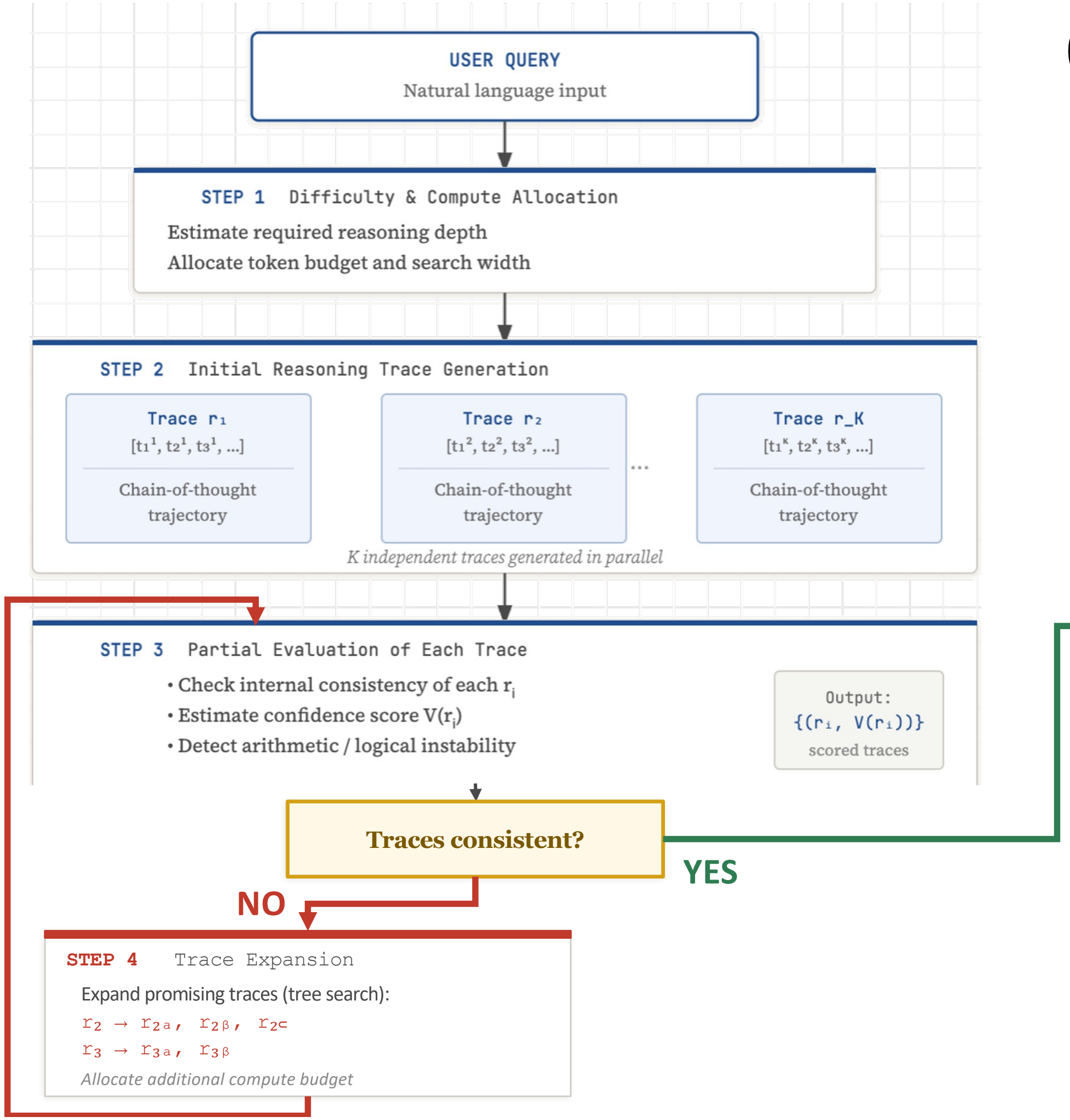


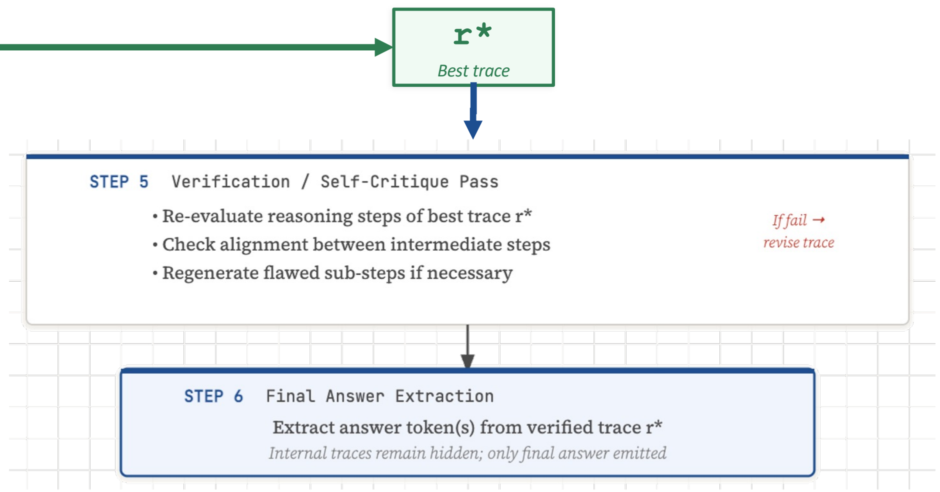
https://arxiv.org/pdf/2408.03314

O1 flow

Instead of producing a short reasoning chain, they:

- Sample multiple reasoning trajectories
- Score or compare them
- Choose the most consistent
- Allocate more tokens to “thinking”
- Delay final answer emission

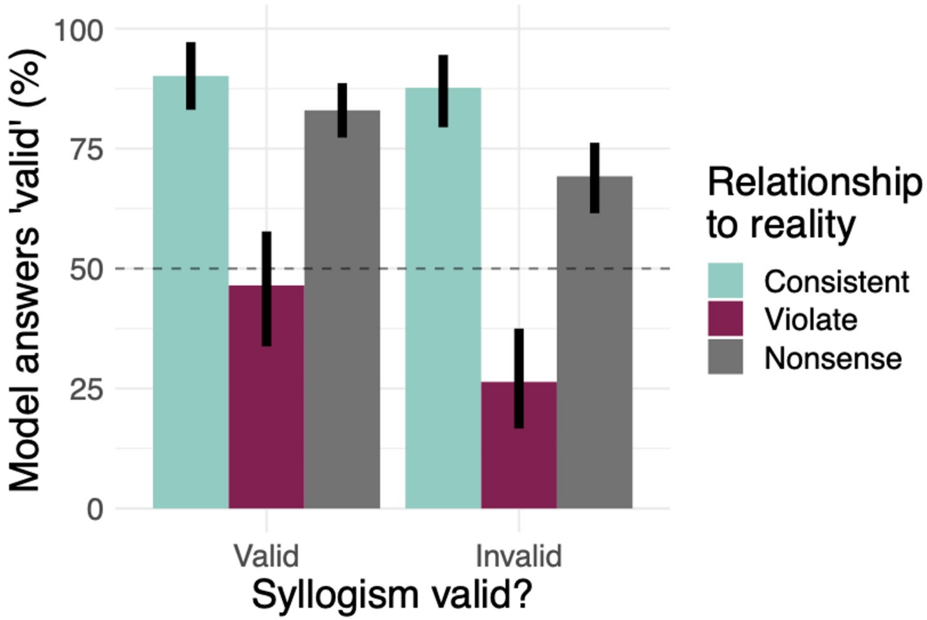




Language models show human-like content effects on reasoning

Ishita Dasgupta[*,1], Andrew K. Lampinen[*,1], Stephanie C. Y. Chan[1], .
James L. McClelland[1,2] and Felix Hill[1]
[*]Equal contributions, listed alphabetically, [1]DeepMind, [2]Stanford University

https://arxiv.org/pdf/2207.07051



	Consistent	Violate	Nonsense
Syllogisms	All **guns** are **weapons**. All **weapons** are **dangerous things**. All **guns** are **dangerous things**.	All **dangerous things** are **weapons**. All **weapons** are **guns**. All **dangerous things** are **guns**.	All **zoct** are **spuff**. All **spuff** are **thrund**. All **zoct** are **thrund**.

LLMs conflate their internal knowledge with given task knowledge …. Sometimes you want the model to only reason with given facts … Sometimes you need guarantees that this actually happens.

Symbolic/Formal Reasoning is a way to achieve this.

Symbolic Representation

- Representing the meaning of natural language is ultimately a difficult philosophical question, i.e. the "meaning of meaning".
- Traditional approach is to map ambiguous NL to unambiguous logic in ***first-order logic*** (***FOL***), also known as predicate logic.
- Standard inference (theorem proving) methods exist for FOL that can determine when one statement entails (implies) another. Questions can be answered by determining what potential responses are entailed by given NL statements and background knowledge all encoded in FOL.

The step step is broadly called Autoformalization.

Autoformalization is the process of automatically translating from natural language to formal specifications and proofs.

https://openreview.net/pdf?id=IUikebJ1Bf0

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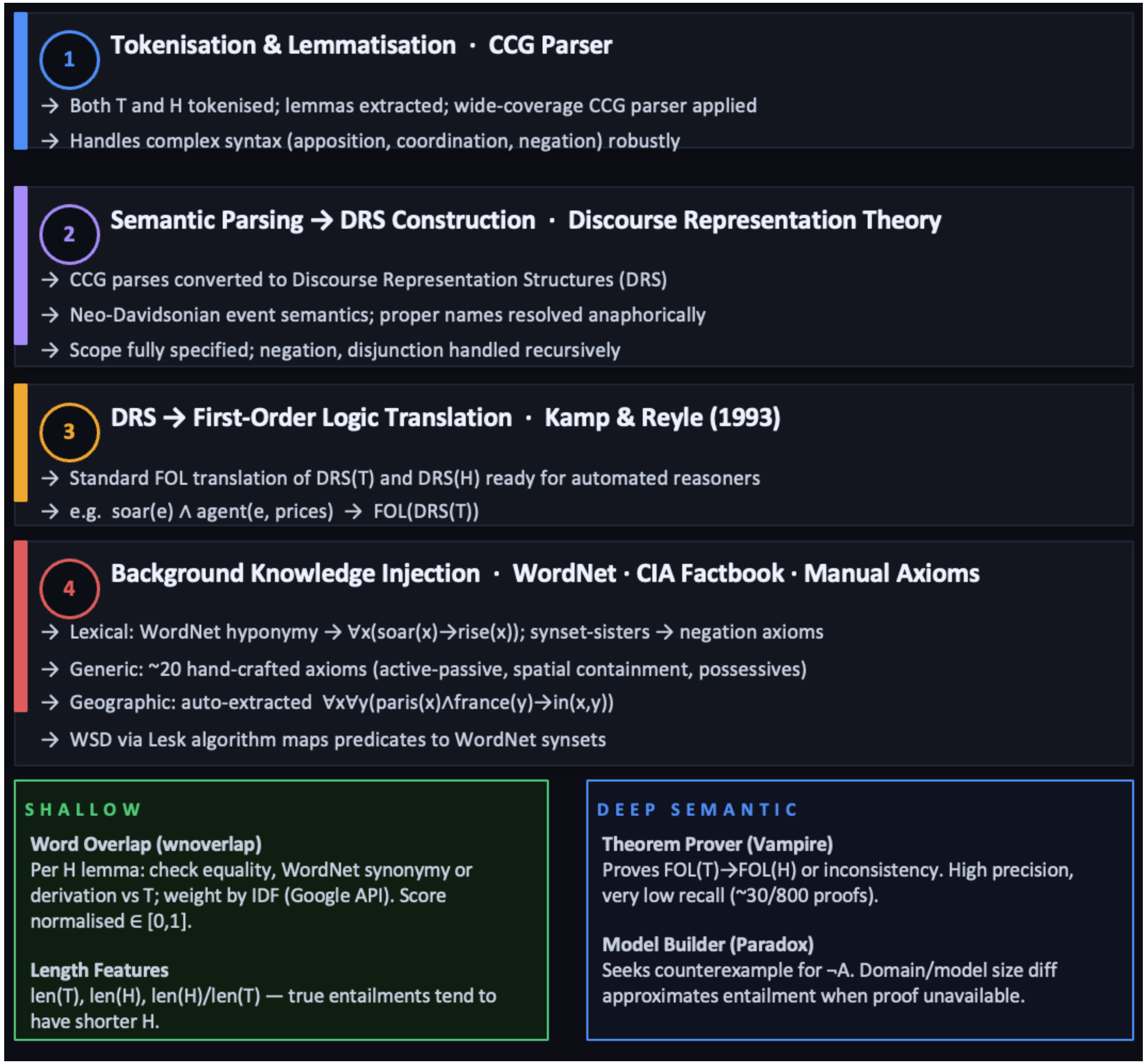
Recognising Textual Entailment

Example: 78 (FALSE)

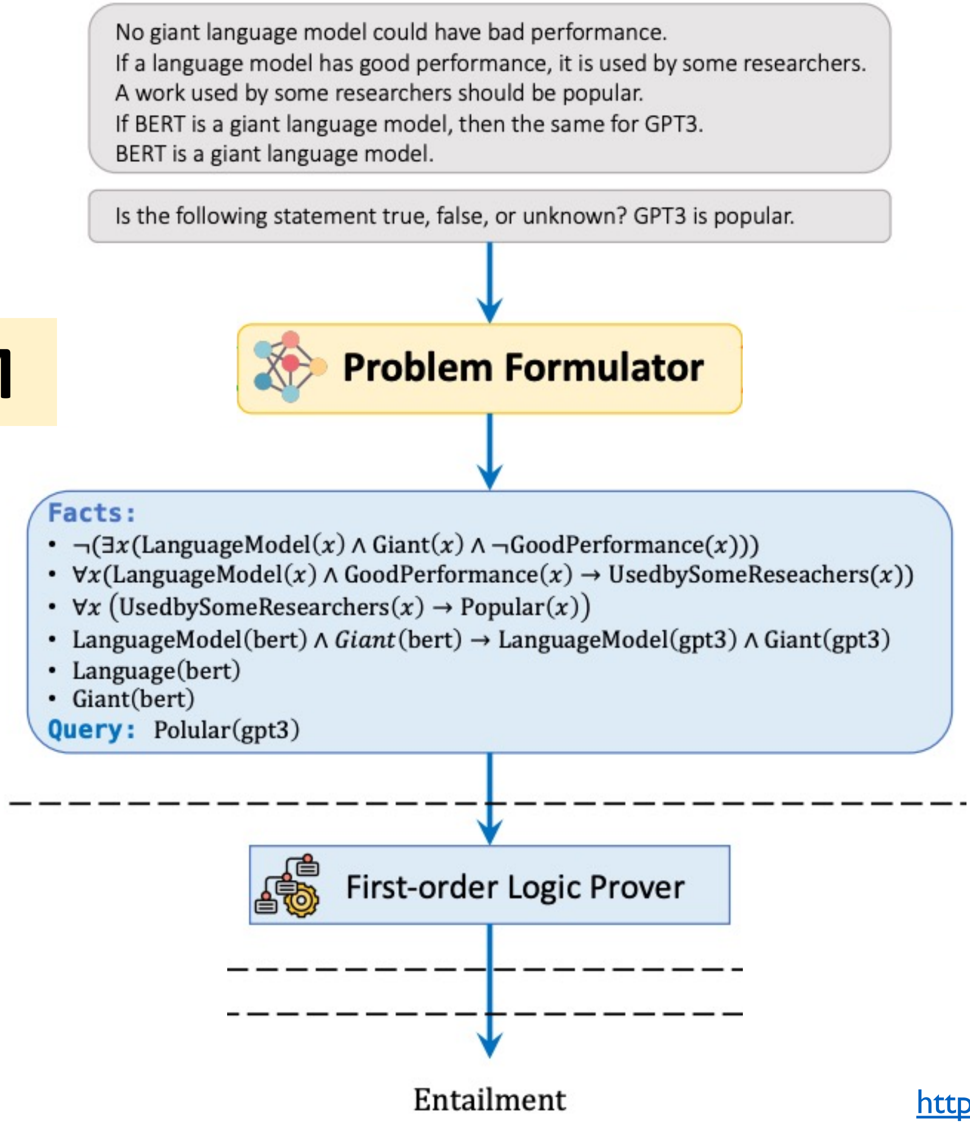
T: Clinton's new book is not big seller here.
H: Clinton's book is a big seller.



https://aclanthology.org/H05-1079.pdf

Same but now with LLMs

LLM



https://arxiv.org/abs/2305.12295